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自制主板 BIOS Port 80 码定义					
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宁畅信息 三级文件	
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1 目的

本文件是 Intel、Hygon、AMD 平台自制主板 BIOS Port80 规格标准，规范的定义，可以减少项目开发，测试、维护过程中因定义不清楚造成的问题。本文档为参考标准，为产品开发过程中的通用做法。具体产品的实现可能因超频需求的不同而有所差异，原则上产品需求中没有特别说明的功能项均按照此标准执行。

2 范围

本过程文件使用范围为宁畅信息公司产品，属于内部文档。

3 职责

本文档适用于 BIOS 开发、BIOS 测试、系统测试、主板维修工程师及项目审查的相关领导。

4 定义

POST: 开机自检（Power On Self Test）的缩写。开机到载入操作系统之前，POST 将检测各种部件。

SEC: Security Phase

DXE: Driver eXecution Environment

PEI: Pre-EFI Initialization

PEIM: Pre-EFI Initialization Modules

SPD: Serial Presence Detect

QPI: Quick Path Interconnect

PPI: PEIM to PEIM Interface

5 Intel 平台 Port 80 码定义信息

Progress Code	Port 0x84 Code	Description	Port 0x85 Code	Description
SEC Phase				
POWER_ON_POST_CODE	0x01	Enter flat 32-bit mode	0x00	
MICROCODE_POST_CODE	0x02	Microcode load begin	0x00	

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NEM_CACHE_INIT_POST_CODE	0x03	Enable NEM mode	0x00	
NEM_CACHE_ENABLED_POST_CODE	0x04	Pei Cache When Disabled	0x00	
SBSP_PBSP_BRANCH_POST_CODE	0x05	SBSP or PBSP Called	0x00	
SEC_CPU_INIT_END_POST_CODE	0x06	Uncore BUS/MCFG configuration	0x00	
NON_INTEL_CPU_ERROR_CODE	0xCA	None of Intel CPU Populated	0x00	
VCU_COMMAND_TIMEOUT_ERROR_CODE	0xCB	Uncore Init Failed		
ILLEGAL_SOCKET_ID_ERROR_CODE	0xCC	Failed to Enable Access to CSRs		
MCU_LOAD_ERROR_CODE	0xCD	Microcode Loading Error	0x00	
MMCFG_NOT_YET_ENABLED_ERROR_CODE	0xCE	MMCFG Enable Yet	0x00	
NEM_NO_FREE_MTRR_ERROR_CODE	0xD0	NEM Data Stack Test Failed	0x00	
NEM_DATA_RW_TEST_ERROR_CODE	0xD1	Configuration Test Failed	0x00	
PEI Phase				
PEI_CORE_STARTED	0x10	PEI Core Started	0x00	
PEI_CAR_CPU_INIT	0x11	CPU PEIM	0x00	
reserved for CPU	0x12-0x14	reserved for CPU		

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PEI_CAR_NB_INIT	0x15	Pre-memory NB Initialization	0x00	
reserved for NB	0x16-0x18	reserved for NB		
PEI_CAR_SB_INIT	0x19	Pre-memory SB Initializatio	0x00	
reserved for SB	0x1A-0x1C	reserved for SB		
reserved for OEM use	0x1D-0x2A	reserved for OEM use		
PEI_MEMORY_SPD_READ	0x2B	Memory Initialization - SPD Read.	0x00	
PEI_MEMORY_PRESENCE_DETECTION	0x2C	Memory presence detection	0x00	
PEI_MEMORY_TIMING	0x2D	Gather Remaining SPD Data	0x00	
PEI_MEMORY_CONFIGURING	0x2E	Train DDR	0x00	
PEI_MEMORY_OPTIMIZING	0xB9	Hardware Memory Test and Initialization		
PEI_MEMORY_INIT	0x2F	Memory Initialization Start	0x00	
reserved for AML use	0x30	reserved for AML use		
PEI_MEMORY_INSTALLED	0x31	Memory Initialization Complete	0x00	
PEI_CPU_INIT	0x32	CPU POST-Memory Initialization	0x00	
PEI_CPU_CACHE_INIT	0x33	CPU Cache initialization	0x00	

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PEI_CUP_AP_INIT	0x34	Application Processor(s) (AP) initialization	0x00	
PEI_CPU_BSP_SELECT	0x35	CPU PEIM (BSP Select)	0x00	BSP Selection
PEI_CPU_SMM_INIT	0x36	CPU PEIM (CPU SMM Init)	0x00	SMM Initialization
PEI_MEM_NB_INIT	0x37	NB PEIM (Mem_NB_Init)	0x00	POST-Memory NB Initialization
reserved for NB	0x38-0x3A	reserved for NB		
PEI_MEM_SB_INIT	0x3B	SB PEIM (Mem_SB_Init)	0x00	POST-Memory SB Initialization
reserved for SB	0x3C-0x3E	reserved for SB		
reserved for OEM use	0x3F-0x4E	reserved for OEM use		
PEI_DXE_IPL_STARTED	0x4F	Dxe IPL started	0x00	
PEI_RECOVERY_AUTO	0xF0	Recovery	0x00	
PEI_RECOVERY_USER	0xF1	PEIM which detected forced Recovery condition	0x00	
PEI_RECOVERY_STARTED	0xF2	Recovery PEIM (Recovery started)	0x00	
PEI_RECOVERY_CAPSULE_FOUN ND	0xF3	Recovery PEIM (Capsule found)	0x00	

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PEI_RECOVERY_CAPSULE_LOA DED	0xF4	Recovery PEIM (Capsule Loaded)	0x00	
PEI_S3_STARTED	0xE0	S3 Resume PEIM (S3 started)	0x00	
PEI_S3_BOOT_SCRIPT	0xE1	S3 Resume PEIM (S3 boot script)	0x00	
PEI_S3_VIDEO_REPOST	0xE2	S3 Resume PEIM (S3 Video Repost)	0x00	
PEI_S3_OS_WAKE	0xE3	S3 Resume PEIM (S3 OS wake)	0x00	
DXE Phase				
DXE_CORE_STARTED	0x60	DXE Core started	0x00	
DXE_NVRAM_INIT	0x61	DXE NVRAM Init	0x00	
DXE_SBRUN_INIT	0x62	SB RUN Init	0x00	
DXE_CPU_INIT	0x63	Dxe CPU Init	0x00	CPU DXE Initialization
0x90800001	0x63	UsbLanInit	0x01	UsbLanInit
reserved for CPU	0x64-0x67	reserved for CPU		
DXE_NB_HB_INIT	0x68	DXE PCI Host Bridge Init	0x00	
DXE_NB_INIT	0x69	DXE NB Init	0x00	
DXE_NB_SMM_INIT	0x6A	DXE NB SMM Init	0x00	

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reserved for NB	0x6B-0x6F	reserved for NB		
DXE_SB_INIT	0x70	DXE SB Init	0x00	DXE SB Init
0x90000001	0x70	DXE SB Init	0x01	Smbios Data Update Dxe
0x90000002	0x70	DXE SB Init	0x02	Pci Bus Dxe
0x90000003	0x70	DXE SB Init	0x03	Redfish Hi Dxe
0x90000004	0x70	DXE SB Init	0x04	Si Cpu Init Dxe
0x90000005	0x70	DXE SB Init	0x05	IIO Init
DXE_SB_SMM_INIT	0x71	DXE SB SMM Init	0x00	
DXE_SB_DEVICES_INIT	0x72	DXE SB devices Init	0x00	
reserved for SB	0x73-0x77	reserved for SB		
DXE_ACPI_INIT	0x78	DXE ACPI Init	0x00	
DXE_CSM_INIT	0x79	DXE CSM Init	0x00	CSM Driver Entry point
0x90100001	0x79	DXE CSM Init	0x01	Pi Smm Ipl
0x90100002	0x79	DXE CSM Init	0x02	Pch Smi Dispatcher
0x90100003	0x79	DXE CSM Init	0x03	Ras Acpi

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0x90100004	0x79	DXE CSM Init	0x04	Pch Init Smm
0x90100005	0x79	DXE CSM Init	0x05	Processor Error Handler
0x90100006	0x79	DXE CSM Init	0x06	Error Control
0x90100007	0x79	DXE CSM Init	0x07	Sps Dxe
0x90100008	0x79	DXE CSM Init	0x08	Cpu Features Dxe
0x90100009	0x79	DXE CSM Init	0x09	Ppm Initialize Dxe
0x9010000A	0x79	DXE CSM Init	0x0A	Socket Setup
0x9010000B	0x79	DXE CSM Init	0x0B	UefiRestNetworkEntryPoint
reserved for AMI use	0x7A-0x7F	reserved for IBV use		
reserved fpr OEM use	0x80-0x8F	reserved fpr OEM use		
DXE_BDS_STARTED	0x90	DXE BDS Started	0x00	
DXE_BDS_CONNECT_DRIVERS	0x91	DXE BDS connect drivers	0x00	
0x90200001	0x91		0x01	BDS Hook dispatch BPS drivers
0x90200002	0x91		0x02	BDS Hook Connect Dcpmm Handles
0x90200003	0x91		0x03	Connect Root Bridge Handles Before Pci Bus Enum

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DXE_PCI_BUS_BEGIN	0x92	DXE PCI Bus begin	0x01	
0x90300001	0x92	DXE PCI Bus begin	0x01	Create Pci Setup Data Buffer Start.
0x90300002	0x92	DXE PCI Bus begin	0x02	Create Pci Setup Data Buffer End.
0x90300003	0x92	DXE PCI Bus begin	0x03	Create Irq Tables Start.
0x90300004	0x92	DXE PCI Bus begin	0x04	Create Irq Tables End.
0x90300005	0x92	DXE PCI Bus begin	0x05	Start Pci Devices Start.
0x90300015	0x92	DXE PCI Bus begin	0x15	InstallPciDevice Start
0x90300016	0x92	DXE PCI Bus begin	0x16	InstallPciDevice End
0x90300017	0x92	DXE PCI Bus begin	0x17	Activate3rdPartyRom Start
0x90300018	0x92	DXE PCI Bus begin	0x18	Activate4rdPartyRom End
0x90300019	0x92	DXE PCI Bus begin	0x19	GetOptRom Start
0x9030001A	0x92	DXE PCI Bus begin	0x1A	GetOptRom End
0x9030001B	0x92	DXE PCI Bus begin	0x1B	ActivateOptRom Start
0x9030001C	0x92	DXE PCI Bus begin	0x1C	ActivateOptRom End
0x9030001D	0x92	DXE PCI Bus begin	0x1D	InstallEfiOpRom Start

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0x9030001E	0x92	DXE PCI Bus begin	0x1E	InstallEfiOpRom End
0x90300006	0x92	DXE PCI Bus begin	0x06	Start Pci Devices End.
0x90300007	0x92	DXE PCI Bus begin	0x07	Init Pci Local Data Start.
0x90300008	0x92	DXE PCI Bus begin	0x08	Init Pci Local Data End.
0x90300009	0x92	DXE PCI Bus begin	0x09	Bus Enumeration Start.
0x9030000A	0x92	DXE PCI Bus begin	0x0A	Bus Enumeration End.
0x9030000B	0x92	DXE PCI Bus begin	0x0B	BDS Update BootOption Variables.
0x9030000C	0x92	DXE PCI Bus begin	0x0C	BDS Adjust Efi Os BootOrder.
0x9030000D	0x92	DXE PCI Bus begin	0x0D	Init Pci Devices Data Start.
0x9030000E	0x92	DXE PCI Bus begin	0x0E	Init Pci Devices Data End.
0x9030000F	0x92	DXE PCI Bus begin	0x0F	Ami Pldm Interface Driver Start.
0x90300010	0x92	DXE PCI Bus begin	0x10	Ami Pldm Interface Driver End.
0x90300011	0x92	DXE PCI Bus begin	0x11	SuperFW Smbios Start.
0x90300012	0x92	DXE PCI Bus begin	0x12	Send Recieve Rest Http Command Start.
0x90300013	0x92	DXE PCI Bus begin	0x13	Send Recieve Rest Http Command End.

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0x90300014	0x92	DXE PCI Bus begin	0x14	Ready To Boot.
DXE_PCI_BUS_HPC_INIT	0x93	DXE PCI Bus HPC Init	0x00	
DXE_PCI_BUS_ENUM	0x94	DXE PCI Bus enumeration	0x00	
DXE_PCI_BUS_REQUEST_RESOURCES	0x95	DXE PCI Bus resource requested	0x00	
0x90400001	0x95	DXE PCI Bus resource requested	0x01	PCI Bus Request Resources End.
DXE_PCI_BUS_ASSIGN_RESOURCES	0x96	DXE PCI Bus assign resource	0x00	
0x90500001	0x96	DXE PCI Bus assign resource	0x01	PCI Bus Assign Resources End.
DXE_CON_OUT_CONNECT	0x97	DXE CON_OUT connect	0x00	
DXE_CON_IN_CONNECT	0x98	DXE CON_IN connect	0x00	
0x90600001	0x98	DXE CON_IN connect	0x01	Virtual Usb Interface Open.
0x90600002	0x98	DXE CON_IN connect	0x02	Redfish Hi Driver Start.
0x90600003	0x98	DXE CON_IN connect	0x03	Redfish Hi Driver End.
0x90600004	0x98	DXE CON_IN connect	0x04	Redfish Hi Check Bmc Start.
0x90600005	0x98	DXE CON_IN connect	0x05	Redfish Hi Check Bmc End.
0x90600006	0x98	DXE CON_IN connect	0x06	BDS Connect Everything Start.

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0x90600007	0x98	DXE CON_IN connect	0x07	BDS Connect Everything End.
0x90600008	0x98	DXE CON_IN connect	0x08	Virtual Usb Interface Close.
DXE_SIO_INIT	0x99	DXE SIO Init	0x00	
DXE_USB_BEGIN	0x9A	DXE USB start	0x00	
DXE_USB_RESET	0x9B	DXE USB reset	0x00	
DXE_USB_DETECT	0x9C	DXE USB detect	0x00	
DXE_USB_ENABLE	0x9D	DXE USB enable	0x00	
reserved for AMI use	0x9E-0x9F	reserved for IBV use		
DXE_IDE_BEGIN	0xA0	DXE IDE begin	0x00	
DXE_IDE_RESET	0xA1	DXE IDE reset	0x00	
DXE_IDE_DETECT	0xA2	DXE IDE detect	0x00	
DXE_IDE_ENABLE	0xA3	DXE IDE enable	0x00	
DXE_SCSI_BEGIN	0xA4	DXE SCSI begin	0x00	
DXE_SCSI_RESET	0xA5	DXE SCSI reset	0x00	
DXE_SCSI_DETECT	0xA6	DXE SCSI detect	0x00	

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DXE_SCSI_ENABLE	0xA7	DXE SCSI enable	0x00	
DXE_SETUP_VERIFYING_PASS WORD	0xA8	DXE verifying SETUP password	0x00	
reserved for AML use	0xAA	reserved for AML use		
DXE_SETUP_INPUT_WAIT	0xAB	DXE SETUP input wait	0x00	
reserved for AML use	0xAC	reserved for AML use		
DXE_READY_TO_BOOT	0xAD	DXE Ready to Boot	0x00	
0x90700001	0xAD	DXE Ready to Boot	0x01	Save Memory Type Information.
0x90700002	0xAD	DXE Ready to Boot	0x02	Exit Boot Services.
0x90700003	0xAD	DXE Ready to Boot	0x03	Run Wdt Before Os Boot.
DXE_LEGACY_BOOT	0xAE	DXE Legacy Boot	0x00	
DXE_EXIT_BOOT_SERVICES	0xAF	DXE Exit Boot Services	0x00	
RT_SET_VIRTUAL_ADDRESS_M AP_BEGIN	0xB0	RT Set Virtual Address Map Begin	0x00	
RT_SET_VIRTUAL_ADDRESS_M AP_END	0xB1	RT Set Virtual Address Map End	0x00	
DXE_LEGACY_OPROM_INIT	0xB2	DXE Legacy Option ROM init	0x00	
DXE_RESET_SYSTEM	0xB3	DXE Reset system	0x00	

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DXE_USB_HOTPLUG	0xB4	DXE USB Hot plug	0x00	
DXE_PCI_BUS_HOTPLUG	0xB5	DXE PCI BUS Hot plug	0x00	
DXE_NVRAM_CLEANUP	0xB6	DXE NVRAM cleanup	0x00	
DXE_CONFIGURATION_RESET	0xB7	DXE Configuration Reset	0x00	
reserved for AMI use	0xB8-0xBF	reserved for AMI use		
reserved for OEM use	0xC0-0xCF	reserved for OEM use		
PEI Error Code				
PEI_MEMORY_INVALID_TYPE	0x50	Memory Type Invalid	0x00	Memory Type Invalid
PEI_MEMORY_INVALID_SPEED	0x50	Memory Speed Invalid	0x01	Memory Speed Invalid
PEI_MEMORY_SPD_FAIL	0x51	Memory SPD failure	0x00	Memory SPD failure
PEI_MEMORY_INVALID_SIZE	0x52	Memory Size Invalid	0x00	Memory Size Invalid
PEI_MEMORY_MISMATCH	0x52	Memory Mismatch	0x01	Memory Mismatch
PEI_MEMORY_NOT_DETECTED	0x53	Memory not detected	0x00	Memory not detected
PEI_MEMORY_NONE_USEFUL	0x53	None of the Memory installed valid	0x01	None of the Memory installed valid
PEI_MEMORY_ERROR	0x54	Memory Error	0x00	Memory Error

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PEI_MEMORY_NOT_INSTALLED	0x55	Memory not installed	0x00	Memory not installed
PEI_CPU_INVALID_TYPE	0x56	CPU type invalid	0x00	CPU type invalid
PEI_CPU_INVALID_SPEED	0x56	CPU Speed invalid	0x01	CPU Speed invalid
PEI_CPU_MISMATCH	0x57	CPU Mismatch	0x00	CPU Mismatch
PEI_CPU_SELF_TEST_FAILED	0x58	PEI CPU Self-test failure	0x00	PEI CPU Self-test failure
PEI_CPU_CACHE_ERROR	0x58	PEI CPU Cache Error	0x01	PEI CPU Cache Error
PEI_CPU_MICROCODE_UPDATE_FAILED	0x59	CPU Microcode Update failure	0x00	CPU Microcode Update failure
PEI_CPU_NO_MICROCODE	0x59	No CPU microcode	0x01	No CPU microcode
PEI_CPU_INTERNAL_ERROR	0x5A	PEI CPU internal error	0x00	PEI CPU internal error
PEI_CPU_ERROR	0x5A	PEI CPU error	0x01	PEI CPU error
PEI_RESET_NOT_AVAILABLE	0x5B	PEI reset not available	0x00	PEI reset not available
reserved for AMI use	0x5C-0x5F	reserved for AMI use		
PEI_RECOVERY_PPI_NOT_FOUND	0xF8	Recover Function Not Found	0x00	
PEI_RECOVERY_NO_CAPSULE	0xF9	Recovery Capsule Not Found	0x00	
PEI_RECOVERY_INVALID_CAPSULE	0xFA	Invalid Recovery Capsule	0x00	

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reserved for AMI use	0xFB-0xFF	reserved for AMI use		
PEI_MEMORY_S3_RESUME_FAILED	0xE8	S3 Resume Failed	0x00	
PEI_S3_RESUME_PPI_NOT_FOUND	0xE9	S3 Resume PPI Not Found	0x00	
PEI_S3_BOOT_SCRIPT_ERROR	0xEA	S3 Resume Boot Script Error	0x00	
PEI_S3_OS_WAKE_ERROR	0xEB	OS S3 Wake Up Error	0x00	
reserved for AMI use	0xEC-0xEF	reserved for AMI use		
DXE Error Code				
DXE_CPU_ERROR	0xD0	DXE CPU initialization error	0x00	
DXE_NB_ERROR	0xD1	DXE NB initialization error	0x00	
DXE_SB_ERROR	0xD2	DXE SB initialization error	0x00	
DXE_ARCH_PROTOCOL_NOT_AVAILABLE	0xD3	Some Architectural Protocols are not available	0x00	
DXE_PCI_BUS_OUT_OF_RESOURCES	0xD4	Out of resources for PCI device	0x00	
DXE_LEGACY_OPROM_NO_SPACE	0xD5	Not enough space for legacy OpROM	0x00	
DXE_NO_CON_OUT	0xD6	No ConOut	0x00	
DXE_NO_CON_IN	0xD7	No ConIn	0x00	

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DXE_INVALID_PASSWORD	0xD8	Invalid password	0x00	
DXE_BOOT_OPTION_LOAD_ERROR	0xD9	Invalid HDD password	0x00	
DXE_BOOT_OPTION_FAILED	0xDA	Boot option failed	0x00	
DXE_FLASH_UPDATE_FAILED	0xDB	Flash update failed	0x00	
DXE_RESET_NOT_AVAILABLE	0xDC	Reset protocol is not available	0x00	
KTI POST code				
STS_DATA_STRUCTURE_INIT/ MINOR_STS_DATA_INIT	0xA0	Initialize KTIRC input structure default values	0x00	
STS_COLLECT_EARLY_SYSTEM_INFO/ MINOR_STS_COLLECT_INFO	0xA1	Collect info such as SBSP, Boot Mode, Reset type etc	0x00	
STS_MDFIS_SWEEP_TRAINING/ MINOR_STS_MDFIS_SWEEP_TRAINING_1	0xA2	Check Point before issue Mdfis Sweep Training	0x00	
STS_MDFIS_SWEEP_TRAINING/ MINOR_STS_MDFIS_SWEEP_TRAINING_2	0xA2	Check Point before issue Mdfis Sweep Training	0x01	
STS_MDFIS_SWEEP_TRAINING/ MINOR_STS_MDFIS_SWEEP_TRAINING_3	0xA2	Check Point before issue Mdfis Sweep Training	0x02	
STS_MDFIS_SWEEP_TRAINING/ MINOR_STS_MDFIS_SWEEP_TRAINING_4	0xA2	Check Point before issue Mdfis Sweep Training	0x03	
STS_MDFIS_SWEEP_TRAINING/ MINOR_STS_MDFIS_SWEEP_TRAINING_5	0xA2	Check Point before issue Mdfis Sweep Training	0x04	
STS_MDFIS_SWEEP_TRAINING/ MINOR_STS_MDFIS_SWEEP_TRAINING_6	0xA2	Check Point before issue Mdfis Sweep Training	0x05	

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STS_SETUP_MINIMUM_PATH/ MINOR_STS_ADD_SOCKET_TO _MIN_PATH_TREE	0xA3	Setup up minimum path between SBSP & other sockets	0x01	
STS_SETUP_MINIMUM_PATH/ MINOR_STS_COLLECT_LEP	0xA3	Setup up minimum path between SBSP & other sockets	0x02	
STS_SETUP_MINIMUM_PATH/ MINOR_STS_CHK_PBSP_CHKIN	0xA3	Setup up minimum path between SBSP & other sockets	0x03	
STS_SETUP_MINIMUM_PATH/ MINOR_STS_SET_PBSP_BOOT_ PATH	0xA3	Setup up minimum path between SBSP & other sockets	0x04	
STS_SETUP_MINIMUM_PATH/ MINOR_STS_SET_S BSP_CONFI G_PATH	0xA3	Setup up minimum path between SBSP & other sockets	0x05	
STS_SETUP_MINIMUM_PATH/ MINOR_STS_REBUILD_MINI_P ATH_TREE	0xA3	Setup up minimum path between SBSP & other sockets	0x06	
STS_PBSP_SYNC_UP/MINOR_S TS_ISSUE_BOOT_GO	0xA6	Sync up with PBSPs	0x01	
STS_TOPOLOGY_DISCOVERY/M INOR_STS_CHK_TOPOLOGY	0xA7	Topology discovery and route calculation	0x01	
STS_TOPOLOGY_DISCOVERY/M INOR_STS_DETECT_RING	0xA7	Topology discovery and route calculation	0x02	
STS_TOPOLOGY_DISCOVERY/M INOR_STS_CONSTRUCT_TOPOL OGY_TREE	0xA7	Topology discovery and route calculation	0x03	
STS_TOPOLOGY_DISCOVERY/M INOR_STS_CALCULATE_ROUTE	0xA7	Topology discovery and route calculation	0x04	
STS_PROGRAM_FINAL_ROUTE /MINOR_STS_PROGRAM_FINA L_ROUTE_START	0xA8	Program final route	0xFF	
STS_PROGRAM_FINAL_ROUTE /MINOR_STS_PROGRAM_FINA L_FPGA_START	0xA8	Program final route	0x01	
STS_PROGRAM_FINAL_IO_SAD /MINOR_STS_ALLOCATE_CPU_ RESOURCE	0xA9	Program final IO SAD setting	0x01	

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STS_PROGRAM_FINAL_IO_SAD /MINOR_STS_FILL_SAD_TGTLST	0xA9	Program final IO SAD setting	0x02	
STS_PROGRAM_FINAL_IO_SAD /MINOR_STS_PROGRAM_CPU_SAD_ENTRIES	0xA9	Program final IO SAD setting	0x03	
STS_PROGRAM_FINAL_IO_SAD /MINOR_STS_EARLY_IO_SAD	0xA9	Program final IO SAD setting	0x04	
STS_PROTOCOL_LAYER_SETTING/MINOR_STS_PROGRAM_RING_CRDT	0xAA	Protocol layer and other Uncore settings	0x01	
STS_PROTOCOL_LAYER_SETTING/MINOR_STS_KTI_MISC_SETUP	0xAA	Protocol layer and other Uncore settings	0x02	
STS_FULL_SPEED_TRANSITION /MINOR_STS_FULL_SPEED_START	0xAB	Transition links to full speed operation	0x00	
STS_FULL_SPEED_TRANSITION /MINOR_STS_CONFIGURATION_PRE_DISPATCHING	0xAB	Transition links to full speed operation	0x01	
STS_FULL_SPEED_TRANSITION /MINOR_STS_CONFIGURATION_POST_DISPATCHING	0xAB	Transition links to full speed operation	0x02	
STS_FULL_SPEED_TRANSITION /MINOR_STS_TXEQ_ADAPTATION_START	0xAB	Transition links to full speed operation	0x03	
STS_FULL_SPEED_TRANSITION /MINOR_STS_TXEQ_ADAPTATION_LOOPBACK	0xAB	Transition links to full speed operation	0x04	
STS_FULL_SPEED_TRANSITION /MINOR_STS_TXEQ_ADAPTATION_SLOW	0xAB	Transition links to full speed operation	0x05	
STS_SYSTEM_COHERENCY_SETUP/MINOR_STS_SYSTEM_COHERENCY_START	0xAE	Coherency Settings	0x00	
STS_KTI_COMPLETE/MINOR_STS_KTI_COMPLETE	0xAF	KTIRC is done	0x00	

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STS_KTI_COMPLETE/MINOR_S TS_UNCORE_MESH_MODE_CO MPLETE	0xAF	KTIRC is done	0x01	
STS_KTI_COMPLETE/MINOR_S TS_UNCORE_PECI_MODE_CO MPLETE	0xAF	KTIRC is done	0x02	
STS_KTI_COMPLETE/MINOR_S TS_SDSI_PPIN_BEFORE_PROGR AM	0xAF	KTIRC is done	0x03	
STS_KTI_COMPLETE/MINOR_S TS_SDSI_PPIN_AFTER_PROGRA M	0xAF	KTIRC is done	0x04	
STS_KTI_COMPLETE/MINOR_S TS_ABOUT_TO_RESET	0xAF	KTIRC is done	0x42	
KTI Error code				
ERR_BOOT_MODE/MINOR_ER R_UNSUPPORTED_BOOT_MOD E	0xD8	err boot mode	0x01	
ERR_BOOT_MODE/MINOR_ER R_PBSP_CHKIN_FAILURE	0xD8	err boot mode	0x02	
ERR_BOOT_MODE/MINOR_ER R_PBSP_ENTER_HLT_FAILURE	0xD8	err boot mode	0x03	
ERR_BOOT_MODE/MINOR_ER R_CPL1_TIMEOUT_FAILURE	0xD8	err boot mode	0x04	
ERR_BOOT_MODE/MINOR_ER R_PBSP_S[X]_BUS_PROGRAM_ FAILURE	0xD8	err boot mode	0x8[X]	X = 0x00 – 0x0F
ERR_MINIMUM_PATH_SETUP/ MINOR_ERR_ADD_SOCKET_TO _TOPOLOGY_TREE	0xD9		0x01	
ERR_MINIMUM_PATH_SETUP/ MINOR_ERR_TOPOLOGY_TREE	0xD9		0x02	
ERR_MINIMUM_PATH_SETUP/ MINOR_ERR_INTERNAL_DATA _STRUCTURE	0xD9		0x03	

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ERR_MINIMUM_PATH_SETUP/ MINOR_ERR_NONCPU_PAREN T_NODE	0xD9		0x04	
ERR_MINIMUM_PATH_SETUP/ MINOR_ERR_INVALID_PORT_C ONNECTION	0xD9		0x05	
ERR_MINIMUM_PATH_SETUP/ MINOR_ERR_SANITY_CHECK	0xD9		0x06	
ERR_MINIMUM_PATH_SETUP/ MINOR_ERR_SLOW_MODE_W A_FAIL	0xD9		0x07	
ERR_TOPOLOGY_DISCOVERY/ MINOR_ERR_ADD_SOCKET_TO _TOPOLOGY_TREE	0xDA		0x01	
ERR_TOPOLOGY_DISCOVERY/ MINOR_ERR_TOPOLOGY_TREE	0xDA		0x02	
ERR_TOPOLOGY_DISCOVERY/ MINOR_ERR_INTERNAL_DATA _STRUCTURE	0xDA		0x03	
ERR_TOPOLOGY_DISCOVERY/ MINOR_ERR_NONCPU_PAREN T_NODE	0xDA		0x04	
ERR_TOPOLOGY_DISCOVERY/ MINOR_ERR_INVALID_PORT_C ONNECTION	0xDA		0x05	
ERR_TOPOLOGY_DISCOVERY/ MINOR_ERR_SANITY_CHECK	0xDA		0x06	
ERR_TOPOLOGY_DISCOVERY/ MINOR_ERR_SLOW_MODE_W A_FAIL	0xDA		0x07	
ERR_SAD_SETUP/MINOR_ERR_ INSUFFICIENT_RESOURCE_SPA CE	0xDB		0x01	
ERR_SAD_SETUP/MINOR_ERR_ INSUFFICIENT_MMIO_L_SPACE	0xDB		0x02	
ERR_SAD_SETUP/MINOR_ERR_ UNEXPECTED_FW_CONDITIO N	0xDB		0x03	

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ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_INVALID_SOCKET_TYPE	0xDC		0x01	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_INVALID_CPU_SOCKET_ID	0xDC		0x02	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_CBO_COUNT_MISMATCH	0xDC		0x03	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_HA_COUNT_MISMATCH	0xDC		0x04	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_M3KTI_COUNT_MISMATCH	0xDC		0x05	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_SKU_MISMATCH	0xDC		0x06	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_LEGACY_PCH_MISMATCH	0xDC		0x07	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_MCP_LINK_COUNT_MISMATCH	0xDC		0x08	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_INVALID_TOPOLOGY	0xDC		0x09	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_TOPOLOGY_CHANGE	0xDC		0x0A	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_INVALID_KTI_PORT_ID	0xDC		0x0B	
ERR_UNSUPPORTED_TOPOLOGY/MINOR_ERR_INVALID_FPGA_TOPOLOGY	0xDC		0x0C	
ERR_FULL_SPEED_TRANSITION/MINOR_ERR_LINK_SPEED_UNSUPPORTED	0xDD		0x01	

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ERR_FULL_SPEED_TRANSITION /MINOR_ERR_MAX_CLK_RATIO_UNSUPPORTED	0xDD		0x02	
ERR_FULL_SPEED_TRANSITION /MINOR_ERR_KTI_ELEC_PARAMETER_NOT_FOUND	0xDD		0x03	
ERR_FULL_SPEED_TRANSITION /MINOR_ERR_KTI_ELEC_PARAMETER_INVALID	0xDD		0x04	
ERR_FULL_SPEED_TRANSITION /MINOR_ERR_S[X]_TIMEOUT_LBC	0xDD		0x8[X]	
ERR_FULL_SPEED_TRANSITION /MINOR_ERR_S[X]_TIMEOUT_IAR	0xDD		0x9[X]	
ERR_S3_RESUME/MINOR_ERR_S3_RESUME_TOPOLOGY_MISMATCH	0xDE		0x01	
ERR_S3_RESUME/MINOR_ERR_S3_RESUME_SETUP_MISMATCH	0xDE		0x02	
ERR_SW_CHECK/XX=0x01 – 0x8F MINOR_ERR_SW_CHECK_XX	0xDF		0xFF	
KTI Warning Code				
WARN_KTI/WARN_LINK_SLOW_SPEED_MODE	0xD7		0x01	
WARN_KTI/WARN_UNSUPPORTED_LINK_SPEED	0xD7		0x02	
WARN_KTI/WARN_LINK_FAILURE	0xD7		0x03	
WARN_KTI/WARN_PER_LINK_OPTION_MISMATCH	0xD7		0x04	
WARN_KTI/WARN_INVALID_LINK	0xD7		0x05	

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WARN_KTI/WARN_ADAPTATION_FAILED	0xD7		0x06	
WARN_KTI/WARN_MMIOH_B ASE_UNSUPPORTED	0xD7		0x09	
WARN_KTI/WARN_MMIOH_S IZE_UNSUPPORTED	0xD7		0x0A	
WARN_KTI/WARN_MMIOH_S IZE_INCOMPATIBLE	0xD7		0x0B	
WARN_KTI/WARN_RESOURCE _NOT_REQUESTED_FOR_CPU	0xD7		0x0C	
WARN_KTI/WARN_RESOURCE _REQUEST_NOT_MET	0xD7		0x0D	
WARN_KTI/WARN_VGA_TARG ET_SOC_NOT_PRESENT	0xD7		0x0E	
WARN_KTI/WARN_VGA_TARG ET_STACK_NOT_PRESENT	0xD7		0x0F	
WARN_KTI/WARN_KTI_MAX_A DDR_UNSUPPORTED	0xD7		0x19	
WARN_KTI/WARN_CPU_STEPP ING_MISMATCH	0xD7		0x10	
WARN_KTI/WARN_TSC_SYNC_ UNSUPPORTED	0xD7		0x11	
Memory Checkpoint				
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_INIT_ CLOCK	0x70		0x00	HBM Init Clock
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_PRE_ TRAINING	0x70		0x01	HBM Pre-Training
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_TRAI NING	0x70		0x08	HBM Training

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CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_MEM _TEST	0x70		0x13	HBM Mem Test
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_MEM _INIT	0x70		0x14	HBM Mem Init
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_SCRA MBLE	0x70		0x15	HBM Setup Scrambling
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_POST _TRAINING	0x70		0x18	HBM Post-Training
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_NOR MAL_MODE	0x70		0x1B	HBM Normal Mode
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_PPR_ FLOW	0x70		0x1C	HBM PPR Flow
CHECKPOINT_MAJOR_HBM/CH ECKPOINT_MINOR_HBM_MBIS T_FLOW	0x70		0x1D	HBM mBIST Flow
CHECKPOINT_MAJOR_HBM_D EBUG	0x71			
CHECKPOINT_MAJOR_HBM_IN TERNAL	0x72			
CHECKPOINT_MAJOR_NVRAM _SOCKET/CHECKPOINT_MINOR _NVRAM_SOCKET_SBSP2AP	0x73		0x00	Nvram socket data SBSP to Aps
CHECKPOINT_MAJOR_PIPE_SY NC/CHECKPOINT_MINOR_PIPE _AP_BOOT_MODE	0x7E		0x00	Pipe Sync AP Boot Mode
CHECKPOINT_MAJOR_PIPE_SY NC/CHECKPOINT_MINOR_PIPE _SBSP_BOOT_MODE	0x7E		0x01	Pipe Sync SBSP Boot Mode
CHECKPOINT_MAJOR_PIPE_SY NC/CHECKPOINT_MINOR_PIPE _AP_DATA	0x7E		0x02	Pipe Sync AP Data

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CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_SBSP_STATUS	0x7E		0x03	Pipe Sync SBSP Status
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_SBSP_DATA	0x7E		0x04	Pipe Sync SBSP Data
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_AP_SSA_DATA	0x7E		0x05	Pipe Sync AP Pre SSA Data
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_SBSP_SSA_DATA	0x7E		0x06	Pipe Sync SBSP Post SSA Data
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_AP_NVRAM_DATA	0x7E		0x07	Pipe Sync AP NVRAM Data
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_SBSP_VAR_DATA	0x7E		0x08	Pipe Sync SBSP Variable Data
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_SBSP_MEM_MODE	0x7E		0x09	Pipe Sync SBSP Memory Mode
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_AP_FINAL_ERROR_SYNC	0x7E		0x0A	Pipe Sync AP Final Error Sync
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_THERMAL_SYNC	0x7E		0x0B	Sync BSP and APs after initThrottlingLate
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_CPL2	0x7E		0x0C	Sync before CPL2 bit is set
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_AP_MEM_BOOT_HEALTH_DATA	0x7E		0x0D	Pipe Sync SBSP Post I/O Heath Check
CHECKPOINT_MAJOR_PIPE_SYNC/CHECKPOINT_MINOR_PIPE_AP_SMBUS_MODE	0x7E		0x0E	Pipe Sync AP Smbus Mode
CHECKPOINT_MAJOR_DIMM_DETECT/CHECKPOINT_MINOR_DIMM_DETECT	0xB0	Detect DIMM population	0x00	

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CHECKPOINT_MAJOR_DIMM_DETECT/CHECKPOINT_MINOR_DISPATCH_SLAVES	0xB0	Detect DIMM population	0x01	
CHECKPOINT_MAJOR_DIMM_DETECT/CHECKPOINT_MINOR_INIT_SMB	0xB0	Detect DIMM population	0x02	
CHECKPOINT_MAJOR_DIMM_DETECT/CHECKPOINT_MINOR_GET_FOLLOWER_DATA	0xB0	Detect DIMM population	0x03	
CHECKPOINT_MAJOR_DIMM_DETECT/CHECKPOINT_MINOR_MAILBOX_READY	0xB0	Detect DIMM population	0x04	
CHECKPOINT_MAJOR_CLOCK_INIT/CHECKPOINT_MINOR_SENDD_FREQ	0xB1	Set DDR frequency	0x01	
CHECKPOINT_MAJOR_CLOCK_INIT/CHECKPOINT_MINOR_SENDD_STATUS	0xB1	Set DDR frequency	0x02	
CHECKPOINT_MAJOR_CLOCK_INIT/CHECKPOINT_MINOR_SET_FREQ	0xB1	Set DDR frequency	0x03	CHECKPOINT_MINOR_SET_FREQ
CHECKPOINT_MAJOR_CLOCK_INIT/CHECKPOINT_MINOR_SET_CLK_VDD	0xB1	Set DDR frequency	0x04	Set Vdd
CHECKPOINT_MAJOR_CLOCK_INIT/CHECKPOINT_MINOR_XMP_INIT	0xB1	Set DDR frequency	0x05	
CHECKPOINT_MAJOR_CLOCK_INIT/CHECKPOINT_MINOR_DCPMM_ENABLE_RESYNC_FSM	0xB1	Set DDR frequency	0x06	
CHECKPOINT_MAJOR_CLOCK_INIT/CHECKPOINT_MINOR_SET_PMIC_VDD	0xB1	Set DDR frequency	0x07	Power on Memory
CHECKPOINT_MAJOR_SPD_DATA	0xB2	Gather remaining SPD data	0x00	Gather remaining SPD data
CHECKPOINT_MAJOR_GLOBAL_EARLY	0xB3		0x00	Early Configuration

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CHECKPOINT_MAJOR_GLOBAL _EARLY/CHECKPOINT_MINOR_ PROG_TIMINGS	0xB3		0x01	
CHECKPOINT_MAJOR_GLOBAL _EARLY/CHECKPOINT_MINOR_ CLOCK_DISABLE	0xB3		0x02	
CHECKPOINT_MAJOR_RANK_D ETECT	0xB4	Evaluate RAS modes and save rank	0x00	Configure DIMM Ranks
CHECKPOINT_MAJOR_RANK_D ETECT/CHECKPOINT_MINOR_S END_DATA	0xB4	Evaluate RAS modes and save rank	0x01	
CHECKPOINT_MAJOR_PARALLE L_DISPATCH	0xB5	—	0x00	Parallel Mode Dispatch
CHECKPOINT_MAJOR_DDRIO_I NIT/CHECKPOINT_MINOR_EAR LY_INIT	0xB6	DDRIO Initialization	0x01	DDRIO Initialization
CHECKPOINT_MAJOR_DDRIO_I NIT/CHECKPOINT_MINOR_LAT E_INIT	0xB6	DDRIO Initialization	0x02	DDRIO Initialization Late
CHECKPOINT_MAJOR_DDRIO_I NIT/CHECKPOINT_MINOR_JED EC_RESET_INIT	0xB6	DDRIO Initialization	0x04	
CHECKPOINT_MAJOR_DDRIO_I NIT/CHECKPOINT_MINOR_DDR T_IO_INIT	0xB6	DDRIO Initialization	0x05	
CHECKPOINT_MAJOR_DDRIO_I NIT/CHECKPOINT_MINOR_TXV REF_CACHE	0xB6	DDRIO Initialization	0x06	
CHECKPOINT_MAJOR_DDRIO_I NIT/CHECKPOINT_MINOR_CLO CK_EN_S3	0xB6	DDRIO Initialization	0x07	
CHECKPOINT_MAJOR_DDRIO_I NIT/CHECKPOINT_MINOR_SEN SE_AMP_CAL	0xB6	DDRIO Initialization	0x08	
CHECKPOINT_MAJOR_DDRIO_I NIT/CHECKPOINT_MINOR_CLO CK_EN	0xB6	DDRIO Initialization	0x09	

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CHECKPOINT_MAJOR_DDRIO_INIT/CHECKPOINT_MINOR_BEF ORE_DDRT_IO_INIT	0xB6	DDRIO Initialization	0x0A	
CHECKPOINT_MAJOR_DDRIO_INIT/CHECKPOINT_MINOR_POWER_STATUS_CHECK	0xB6	DDRIO Initialization	0x0B	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RECEIVE_ENABLE_BASIC	0xB7	Train DDR	0x00	Receive Enable Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_WRITE_LEVELING_BASIC	0xB7	Train DDR	0x01	Write Leveling Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_WR_FLY_BY	0xB7	Train DDR	0x02	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_DQ_DQS_BASIC	0xB7	Train DDR	0x03	Read Dq Dqs: Coarse Read DQ/DQS
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TX_DQ_BASIC	0xB7	Train DDR	0x04	Write Dq Dqs: Coarse Write DQ/DQS
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_LATE_CMD_CLK	0xB7	Train DDR	0x05	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TRAIN_RD_DQS	0xB7	Train DDR	0x06	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TRAIN_WR_DQS	0xB7	Train DDR	0x07	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TRAIN_RD_VREF	0xB7	Train DDR	0x08	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TRAIN_WR_VREF	0xB7	Train DDR	0x09	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_CROSSOVER_CALIBRATION	0xB7	Train DDR	0x0A	

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_SENSE_AMP	0xB7	Train DDR	0x0B	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_CMD_CLK	0xB7	Train DDR	0x0C	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PER_BIT_DESKEW_RX	0xB7	Train DDR	0x0D	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PER_BIT_DESKEW_TX	0xB7	Train DDR	0x0E	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_VREF_TRAINING	0xB7	Train DDR	0x0F	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RMT	0xB7	Train DDR	0x10	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_END_TRAINING	0xB7	Train DDR	0x11	Late Configuration
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_NORMAL_MODE	0xB7	Train DDR	0x12	Switch to Normal Mode
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_ROUND_TRIP_OPTIMIZE	0xB7	Train DDR	0x13	Roundtrip Latency Optimization
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_BACKSIDE_PHASE_TRAINING	0xB7	Train DDR	0x14	MDQS Receive Enable Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_BACKSIDE_CYCLE_TRAINING	0xB7	Train DDR	0x15	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_BACKSIDE_DELAY_TRAINING	0xB7	Train DDR	0x16	MDQS Read Delay Training

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TX_BACKSIDE_FINE_WL_TRAINING	0xB7	Train DDR	0x17	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TX_BACKSIDE_COARSE_WL_TRAINING	0xB7	Train DDR	0x18	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DISABLE_SENSE_AMP	0xB7	Train DDR	0x19	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_CTL_CLK	0xB7	Train DDR	0x1A	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TX_EQ_TRAINING	0xB7	Train DDR	0x1B	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TX_BACKSIDE_DELAY_TRAINING	0xB7	Train DDR	0x1C	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_IMODE_TRAINING	0xB7	Train DDR	0x1D	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TRAIN_WR_VREF_LRDIMM	0xB7	Train DDR	0x1E	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DATA_CONFIGURATION	0xB7	Train DDR	0x1F	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_CMD_VREF_CENTERING	0xB7	Train DDR	0x24	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DQ_SWIZZLE_DISCOVERY	0xB7	Train DDR	0x25	Read Dq Dqs: Swizzle Discovery
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_RID_FINE	0xB7	Train DDR	0x26	

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_RID_COARSE	0xB7	Train DDR	0x27	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_RID_FINAL	0xB7	Train DDR	0x28	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_MEM_TEST	0xB7	Train DDR	0x2B	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PACKAGE_DELAY_CCC	0xB7	Train DDR	0x2C	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_POST_PACKAGE_DELAY_CCC	0xB7	Train DDR	0x2D	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_E_TRAIN_RD_VREF	0xB7	Train DDR	0x31	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_E_TRAIN_WR_VREF	0xB7	Train DDR	0x32	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TRAIN_RD_VREF_LRDIMM	0xB7	Train DDR	0x33	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TX_LRDIMM_DQ_CENTERING	0xB7	Train DDR	0x34	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_LRDIMM_DQ_CENTERING	0xB7	Train DDR	0x35	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DEF_PATH_FINDING	0xB7	Train DDR	0x36	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_WRITE_DEF	0xB7	Train DDR	0x37	Write Dq Dqs: Write DFE Training

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_READ_DEF	0xB7	Train DDR	0x38	Read Dq Dqs: Read DFE Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_BACKSIDE_SWIZZLE	0xB7	Train DDR	0x39	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_CLK_TRAINING	0xB7	Train DDR	0x3A	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_QX_CA_CLK	0xB7	Train DDR	0x3B	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PXC_TRAINING	0xB7	Train DDR	0x3C	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RCSIM_CACHE_DDRIO	0xB7	Train DDR	0x3D	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DQS_SWIZZLE_DISCOVERY	0xB7	Train DDR	0x3E	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PERIODIC_TX_RETRAIN	0xB7	Train DDR	0x3F	Initialize Tx DQ Periodic Retraining
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_FLOW	0xB7	Train DDR	0x40	PPR Flow
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_PRE_TEST	0xB7	Train DDR	0x41	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_PREA_BEFORE	0xB7	Train DDR	0x42	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_DIS_DB_SNOOP	0xB7	Train DDR	0x43	

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_EN_PPR_MODE	0xB7	Train DDR	0x44	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_EN_GUARD_KEY0	0xB7	Train DDR	0x45	
CHECKPOINT_MINOR_PPR_SEQ_EN_GUARD_KEY1	0xB7	Train DDR	0x46	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_EN_GUARD_KEY2	0xB7	Train DDR	0x47	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_EN_GUARD_KEY3	0xB7	Train DDR	0x48	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_WR_PATTERN	0xB7	Train DDR	0x49	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_PREA_AFTER	0xB7	Train DDR	0x4A	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_DIS_PPR_MODE	0xB7	Train DDR	0x4B	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_EN_DB_SNOOP	0xB7	Train DDR	0x4C	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_SEQ_RESTORE_MRO	0xB7	Train DDR	0x4D	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_POST_TEST	0xB7	Train DDR	0x4E	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PPR_POST_TEST	0xB7	Train DDR	0x50	

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NOR_RECEIVE_ENABLE_COARSE				
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RECEIVE_ENABLE_FINE	0xB7	Train DDR	0x51	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RCD_DCA_TIMING_SIMPLE	0xB7	Train DDR	0x52	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RCD_DCA_VREF_SIMPLE	0xB7	Train DDR	0x53	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RCD_DCA_TIMING_COMPLEX	0xB7	Train DDR	0x54	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RCD_DCA_VREF_COMPLEX	0xB7	Train DDR	0x55	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DFE_READ_CENTERING	0xB7	Train DDR	0x56	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DFE_WRITE_CENTERING	0xB7	Train DDR	0x57	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EXTERNAL_WRITE_LEVEL	0xB7	Train DDR	0x58	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_INTERNAL_WRITE_LEVEL	0xB7	Train DDR	0x59	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_LRDIMM_EXTERNAL_WRITE_LEVEL_COARSE	0xB7	Train DDR	0x60	Buffer DRAM Write Leveling Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR	0xB7	Train DDR	0x61	

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NOR_LRDIMM_EXTERNAL_WRITE_LEVEL_FINE				
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_LRDIMM_INTERNAL_WRITE_LEVEL_COARSE	0xB7	Train DDR	0x62	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_LRDIMM_INTERNAL_WRITE_LEVEL_FINE	0xB7	Train DDR	0x63	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_DRAM_RX_EQ	0xB7	Train DDR	0x64	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_TX_EQ	0xB7	Train DDR	0x65	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_IMODE	0xB7	Train DDR	0x66	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_CTL	0xB7	Train DDR	0x67	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_MC_ROM	0xB7	Train DDR	0x68	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_RTT_WR	0xB7	Train DDR	0x69	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_MC_ODT	0xB7	Train DDR	0x6A	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_NT_ODT	0xB7	Train DDR	0x6B	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_DRAM_ROM	0xB7	Train DDR	0x6C	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_DRAM_ROM	0xB7	Train DDR	0x6D	

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NOR_T_RX_ODT				
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TCO_COMP	0xB7	Train DDR	0x6E	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TX_RISE_FALL_SLEW_RATE	0xB7	Train DDR	0x6F	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_CMD_TX_EQ	0xB7	Train DDR	0x70	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DDJC_TRAINING	0xB7	Train DDR	0x71	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_RX_DFE	0xB7	Train DDR	0x72	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_CS_CLK	0xB7	Train DDR	0x73	Host CS Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_CS_CLK_BACKSIDE	0xB7	Train DDR	0x74	RCD QCS Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_CA_CLK	0xB7	Train DDR	0x75	Host CA Training Simple
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_CA_CLK_BACKSIDE	0xB7	Train DDR	0x76	RCD QCA Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_WRITE_LEVELING_BASIC_DDR5	0xB7	Train DDR	0x77	Write Leveling Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PRE_TRAINING_INIT	0xB7	Train DDR	0x78	

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RCOMP_STAT_LEG	0xB7	Train DDR	0x79	Pre-Training Initialization
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DISPLAY_RESULTS	0xB7	Train DDR	0x7A	Display Training Results
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_POST_TRAINING_INIT	0xB7	Train DDR	0x7B	Post-Training Initialization
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_BIT_SWIZZLE_INSPECTION	0xB7	Train DDR	0x7C	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_BCOM_CLK	0xB7	Train DDR	0x7D	BCOM Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PBA_ENUMERATE_ID	0xB7	Train DDR	0x7E	PBA Enumeration
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_T_TX_ECHO	0xB7	Train DDR	0x7F	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_LEGACY_RXDFE	0xB7	Train DDR	0x80	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_REQ_CLK	0xB7	Train DDR	0x81	REQ Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RCD_DCA_DCK_DUTY_CYCLE_TRAINING	0xB7	Train DDR	0x83	RCD DCA/DCK Duty Cycle Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_WRITE_DQDQS_PRE_DFE_2D_CENTERING	0xB7	Train DDR	0x84	Write Dq Dqs: Pre-DFE Write 2D Centering
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_WRITE_DQDQS_POST_DFE_2D_CENTERING	0xB7	Train DDR	0x85	Write Dq Dqs: Post-DFE Write 2D Centering

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_CA_SLEW_RATE	0xB7	Train DDR	0x86	RCD DCA Slew Rate Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DCA_DFE	0xB7	Train DDR	0x87	RCD DCA DFE Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DDRT2_MODE	0xB7	Train DDR	0x8A	Switch to DDRT2 Mode
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_READ_DQDQS_PRE_DFE_2D_CENTERING	0xB7	Train DDR	0x8B	Read Dq Dqs: Pre-DFE Read 2D Centering
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_READ_DQDQS_POST_DFE_2D_CENTERING	0xB7	Train DDR	0x8C	Read Dq Dqs: Post-DFE Read 2D Centering
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_FRONTSIDE_SWIZZLE	0xB7	Train DDR	0x8D	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_BACKSIDE_RX_PER_BIT_DESKEW	0xB7	Train DDR	0x8E	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DCA_TCO	0xB7	Train DDR	0x8F	RCD DCA TCO Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TURNAROUND	0xB7	Train DDR	0x90	Turnarounds Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_LRDIMM_EXTERNAL_WRITE_DELAY	0xB7	Train DDR	0x91	Buffer Write Delay Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DQ_SLEW_RATE	0xB7	Train DDR	0x92	Write Dq Dqs: Slew Rate DQ

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_NORMALIZE_CMD	0xB7	Train DDR	0x93	Command Normalization
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_TCO_DQ	0xB7	Train DDR	0x94	Write Dq Dqs: TCO DQ
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DRAM_DUTY_CYCLE_ADJUSTER	0xB7	Train DDR	0x95	Read Dq Dqs: Duty Cycle Adjuster
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_WRITE_DB_DFE	0xB7	Train DDR	0x96	Buffer Write DFE Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_WRITE_ODT_LATENCY	0xB7	Train DDR	0x97	Write Dq Dqs: Write ODT Latency Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_MCR_EARLY_SWIZZLE_DISCOVERY	0xB7	Train DDR	0x98	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_CTE_INIT_PRE_TRAINING	0xB7	Train DDR	0x9A	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_XTALK_CANCELLATION	0xB7	Train DDR	0x9B	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_LDO_TRAINING	0xB7	Train DDR	0x9C	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RX_JITTER_CANCELLATION	0xB7	Train DDR	0x9D	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_SWEEP_TEST	0xB7	Train DDR	0x9E	MemSweepTester
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_RCD_DCS_DFE	0xB7	Train DDR	0x9F	

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_QCS_SAVE_RESTORE	0xB7	Train DDR	0xA0	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_QCA_SAVE_RESTORE	0xB7	Train DDR	0xA1	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_PERIODIC_RX_RETRAIN	0xB7	Train DDR	0xA2	Initialize Rx DQS Periodic Retraining
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_CA_CLK_COMPLEX	0xB7	Train DDR	0xA3	Host CA Training Complex
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_EARLY_CA_CLK_COMPLEX_RECENTER	0xB7	Train DDR	0xA4	Re-center Host CA Training
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_MRE_SAVE_RESTORE	0xB7	Train DDR	0xA5	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_MRD_SAVE_RESTORE	0xB7	Train DDR	0xA6	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_DWL_SAVE_RESTORE	0xB7	Train DDR	0xA7	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_MWD_SAVE_RESTORE	0xB7	Train DDR	0xA8	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_CHECK_TRAINING_RESULTS	0xB7	Train DDR	0xA9	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_CPGC_DEBUG	0xB7	Train DDR	0xAB	
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_READ_DQDQS_POST_DFE_2D_LATE	0xB7	Train DDR	0xAC	Read Dq Dqs: Post-DFE Read 2D Centering Late

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CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_ENABLE_RX_RETRAINING	0xB7	Train DDR	0xAD	Enable RX Retraining
CHECKPOINT_MAJOR_CHANNEL_TRAINING/CHECKPOINT_MINOR_ENABLE_TX_RETRAINING	0xB7	Train DDR	0xAE	Enable TX Retraining
CHECKPOINT_MAJOR_INIT_THROTTLING/CHECKPOINT_MINOR_EARLY_DDR_THERM	0xB8	Initialize CLTT/OLTT	0x01	Initialize Throttling Early
CHECKPOINT_MAJOR_INIT_THROTTLING/CHECKPOINT_MINOR_MID_DDR_THERM	0xB8	Initialize CLTT/OLTT	0x02	Initialize Throttling
CHECKPOINT_MAJOR_INIT_THROTTLING/CHECKPOINT_MINOR_LATE_DDR_THERM	0xB8	Initialize CLTT/OLTT	0x03	
CHECKPOINT_MAJOR_MEMBIT/CHECKPOINT_MINOR_MSTEP[XX]	0xB9	Hardware memory test and init	0xFF	XX = 0x01 – 0x17
CHECKPOINT_MAJOR_MEMORY_INIT	0xBA	Execute memory init	0x00	Execute memory init
CHECKPOINT_MAJOR_DDR_MEMORY_MAP	0xBB	Program memory map and interleaving	0x00	Initialize Memory Map
CHECKPOINT_MAJOR_DDR_MEMORY_MAP/CHECKPOINT_MINOR_SAD_INTERLEAVE	0xBB	Program memory map and interleaving	0x01	
CHECKPOINT_MAJOR_DDR_MEMORY_MAP/CHECKPOINT_MINOR_TAD_INTERLEAVE	0xBB	Program memory map and interleaving	0x02	
CHECKPOINT_MAJOR_DDR_MEMORY_MAP/CHECKPOINT_MINOR_SAD_NONINTER	0xBB	Program memory map and interleaving	0x03	
CHECKPOINT_MAJOR_DDR_MEMORY_MAP/CHECKPOINT_MINOR_TAD_NONINTER	0xBB	Program memory map and interleaving	0x04	
CHECKPOINT_MAJOR_DDR_MEMORY_MAP/CHECKPOINT_MINOR_RANK_INTER	0xBB	Program memory map and interleaving	0x05	

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CHECKPOINT_MAJOR_DDR_M EMMAP/CHECKPOINT_MINOR _WRITE_SAD	0xBB	Program memory map and interleaving	0x06	
CHECKPOINT_MAJOR_DDR_M EMMAP/CHECKPOINT_MINOR _WRITE_TAD	0xBB	Program memory map and interleaving	0x07	
CHECKPOINT_MAJOR_DDR_M EMMAP/CHECKPOINT_MINOR _WRITE_RIR	0xBB	Program memory map and interleaving	0x08	
CHECKPOINT_MAJOR_DDR_M EMMAP/CHECKPOINT_MINOR _WRITE_SPARE_RANK	0xBB	Program memory map and interleaving	0x0A	
CHECKPOINT_MAJOR_RAS_CO NFIG/CHECKPOINT_MINOR_MI RROR_MODE	0xBC	Program RAS configuration	0x01	
CHECKPOINT_MAJOR_RAS_CO NFIG/CHECKPOINT_MINOR_SP ARE_MODE	0xBC	Program RAS configuration	0x02	
CHECKPOINT_MAJOR_RAS_CO NFIG/CHECKPOINT_MINOR_DE VICE_TAG	0xBC	Program RAS configuration	0x03	
CHECKPOINT_MAJOR_RAS_CO NFIG/CHECKPOINT_MINOR_ER R_THRESH	0xBC	Program RAS configuration	0x04	
CHECKPOINT_MAJOR_RAS_CO NFIG/CHECKPOINT_MINOR_LO CKSTEP_MODE	0xBC	Program RAS configuration	0x05	
CHECKPOINT_MAJOR_RAS_CO NFIG/CHECKPOINT_MINOR_AD DDC_MIRROR	0xBC	Program RAS configuration	0x06	
CHECKPOINT_MAJOR_GET_MA RGINS	0xBD	Rank margin tool		
CHECKPOINT_MAJOR_SSA_API _INIT/CHECKPOINT_MINOR_SS A_API_INIT_START	0xBE		0x01	Rank Margin Test
CHECKPOINT_MAJOR_SSA_API _INIT/CHECKPOINT_MINOR_SS A_API_INIT_EV_LOADER	0xBE		0x02	

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CHECKPOINT_MAJOR_MRC_STOP/ART_STOP/CHECKPOINT_MINOR_MRC_STOP	0xBF	MRC	0x00	
CHECKPOINT_MAJOR_MRC_STOP/ART_STOP/CHECKPOINT_MINOR_MRC_START	0xBF	MRC	0x01	
CHECKPOINT_MAJOR_JEDEC_INIT	0xC0		0x00	
CHECKPOINT_MAJOR_CHECK_POR	0xC1		0x00	Check POR Compatibility
CHECKPOINT_MAJOR_UNLOCK_MEM_REGS	0xC2		0x00	Check Status
CHECKPOINT_MAJOR_CHECK_STATUS	0xC3		0x00	
CHECKPOINT_MAJOR_CONFIG_XMP	0xC4		0x00	
CHECKPOINT_MAJOR_EARLY_INIT_MEM	0xC5		0x00	Initialize Memory
CHECKPOINT_MAJOR_DIMM_INFO/CHECKPOINT_MINOR_DIMM_INFO_EARLY	0xC6		0x01	Socket DIMM Information
CHECKPOINT_MAJOR_DIMM_INFO/CHECKPOINT_MINOR_DIMM_INFO_LATE	0xC6		0x02	DIMM Information After MRC
CHECKPOINT_MAJOR_NVDIMM	0xC7			
CHECKPOINT_MAJOR_SVL_SCRAMBLE	0xC9		0x00	Setup SVL and Scrambling
CHECKPOINT_MAJOR_CMI_CREDIT/CHECKPOINT_MINOR_CMI_CREDIT_INIT	0xCA		0x00	Init CMI Credit Programming
CHECKPOINT_MAJOR_CMI_CREDIT/CHECKPOINT_MINOR_CMI_SET_DONE_BITS	0xCA		0x01	

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CHECKPOINT_MAJOR_CHECK_RAS	0xCB		0x00	Check Ras Support After MemInit
CHECKPOINT_MAJOR_INIT_ADDR	0xCC		0x00	Initialize ADR
CHECKPOINT_MAJOR_INIT_STRUCTURE_LATE	0xCD		0x00	Init Structures Late
CHECKPOINT_MAJOR_MEMORY_LATE	0xCE		0x00	Memory Late
CHECKPOINT_MAJOR_SELECT_BOOT_MODE	0xCF		0x00	Select Boot Mode
CHECKPOINT_MAJOR_MKTME_EARLY_FLOW/CHECKPOINT_MINOR_TME_INIT_FLOW	0xD0		0x00	TME Init Flow
CHECKPOINT_MINOR_MKTME_ENABLE_FLOW/CHECKPOINT_MINOR_TME_PROGRAM_MSRS	0xD0		0x01	TME Flow - PROGRAM_MSRS
CHECKPOINT_MINOR_MKTME_ENABLE_FLOW/CHECKPOINT_MINOR_MKTME_CHECK_CAP	0xD0		0x00	MK-TME Flow
CHECKPOINT_MAJOR_SGX_PREMEM/CHECKPOINT_MINOR_SGX_CHECK_CAP_ST	0xD1		0x00	SGX PreMem Check Capability ST
CHECKPOINT_MAJOR_SGX_PREMEM/CHECKPOINT_MINOR_SGX_CHECK_CAP_MT	0xD1		0x01	SGX PreMem Check Capability MT
CHECKPOINT_MAJOR_SGX_PREMEM/CHECKPOINT_MINOR_SGX_PREMEM	0xD1		0x02	SGX PreMem Init
CHECKPOINT_MAJOR_MEMORY_HEALTH_TEST/CHECKPOINT_MINOR_RUN_MEMORY_HEALTH_TEST	0xD2		0x01	Perform I/O Health Check
CHECKPOINT_MAJOR_MEMORY_HEALTH_TEST/CHECKPOINT_MINOR_CHECK_MEMORY_HEALTH_STATUS	0xD2		0x02	Check I/O Health Check Status

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CHECKPOINT_MAJOR_ENABLE _2N_MODE	0xD3			
CHECKPOINT_MAJOR_CPL2	0xD5		0x00	
CHECKPOINT_MAJOR_OFFSET_ TRAIN_RESULT	0xD6		0x00	Offset training results
CHECKPOINT_MAJOR_DIMM_ MANIFEST/CHECKPOINT_MINO R_DIMM_MANIFEST_PUBLISH	0xD7		0x00	Publish ACTM DIMM Manifest
CHECKPOINT_MAJOR_DIMM_ MANIFEST/CHECKPOINT_MINO R_DIMM_MANIFEST_MSR_WR ITE	0xD7		0x01	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWSG_BEFORE	0xD8		0x00	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWSG_BEFORE_ TEST	0xD8		0x01	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWSG_AFTER_T EST	0xD8		0x02	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWSG_AFTER	0xD8		0x03	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWSR_BEFORE	0xD8		0x04	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWSR_BEFORE_ TEST	0xD8		0x05	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWSR_AFTER_T EST	0xD8		0x06	

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CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWSR_AFTER	0xD8		0x07	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDR_BEFORE	0xD8		0x08	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDR_BEFORE_ TEST	0xD8		0x09	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDR_AFTER_TE ST	0xD8		0x0A	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDR_AFTER	0xD8		0x0B	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDR_BEFORE	0xD8		0x0C	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDR_BEFORE_ TEST	0xD8		0x0D	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDR_AFTER_T EST	0xD8		0x0E	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDR_AFTER	0xD8		0x0F	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDR_BEFORE	0xD8		0x10	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDR_BEFORE_ TEST	0xD8		0x11	

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CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDR_AFTER_T EST	0xD8		0x12	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDR_AFTER	0xD8		0x13	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDR_BEFORE	0xD8		0x14	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDR_BEFORE _TEST	0xD8		0x15	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDR_AFTER_ TEST	0xD8		0x16	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDR_AFTER	0xD8		0x17	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDD_BEFORE	0xD8		0x18	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDD_BEFORE_ TEST	0xD8		0x19	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDD_AFTER_T EST	0xD8		0x1A	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDD_AFTER	0xD8		0x1B	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDD_BEFORE	0xD8		0x1C	

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CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDD_BEFORE _TEST	0xD8		0x1D	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDD_AFTER_T EST	0xD8		0x1E	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDD_AFTER	0xD8		0x1F	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDD_BEFORE	0xD8		0x20	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDD_BEFORE _TEST	0xD8		0x21	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDD_AFTER_T EST	0xD8		0x22	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDD_AFTER	0xD8		0x23	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDD_BEFORE	0xD8		0x24	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDD_BEFORE _TEST	0xD8		0x25	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDD_AFTER_ TEST	0xD8		0x26	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDD_AFTER	0xD8		0x27	

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CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDS_BEFORE	0xD8		0x28	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDS_BEFORE_ TEST	0xD8		0x29	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDS_AFTER_TE ST	0xD8		0x2A	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RRDS_AFTER	0xD8		0x2B	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDS_BEFORE	0xD8		0x2C	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDS_BEFORE_ TEST	0xD8		0x2D	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDS_AFTER_T EST	0xD8		0x2E	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_RWDS_AFTER	0xD8		0x2F	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDS_BEFORE	0xD8		0x30	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDS_BEFORE_ TEST	0xD8		0x31	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WRDS_AFTER_T EST	0xD8		0x32	

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CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDS_AFTER	0xD8		0x33	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDS_BEFORE	0xD8		0x34	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDS_BEFORE _TEST	0xD8		0x35	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDS_AFTER_ TEST	0xD8		0x36	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WWDS_AFTER	0xD8		0x37	
CHECKPOINT_MAJOR_TURNAR OUND/CHECKPIONT_MINOR_T URNAROUND_WA_AFTER	0xD8		0x38	
CHECKPOINT_MAJOR_NO_CO DE	0xFF			
Memory Error Code				
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 000	0xE0		0x00	DeviceType - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 001	0xE0		0x01	DramDieCount - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 002	0xE0		0x02	NumColBits - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 003	0xE0		0x03	NumRowBits - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 004	0xE0		0x04	BankGroupBits - SPD read Byte failed

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ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 005	0xE0		0x05	BankAddressBits - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 006	0xE0		0x06	RcdVendor - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 007	0xE0		0x07	RcdRevision - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 008	0xE0		0x08	DramCapacity - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 009	0xE0		0x09	DramIOWidth - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 010	0xE0		0x0A	RankMixIndicator - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 011	0xE0		0x0B	PackageRanks - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 012	0xE0		0x0C	MiniCycleTime - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 013	0xE0		0x0D	BaseModuleType - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 014	0xE0		0x0E	OperableEndurant VDD - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 015	0xE0		0x0F	RawCardRefDesign - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 016	0xE0		0x10	RawCardRefDesignRev - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 017	0xE0		0x11	NominalHeight - SPD read Byte failed

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ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 018	0xE0		0x12	PrimaryBusWidth - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 019	0xE0		0x13	BusWidthExtension - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 020	0xE0		0x14	ChannelNumPerDimm - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 021	0xE0		0x15	RcdDeviceType - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 022	0xE0		0x16	DramStepping - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 023	0xE0		0x17	PartNum - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 024	0xE0		0x18	SpecRevision - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 025	0xE0		0x19	ManufacturerId - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 026	0xE0		0x1A	TrpVal - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 027	0xE0		0x1B	Module SerialNumber - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 028	0xE0		0x1C	ManufacturingDate - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 029	0xE0		0x1D	TrasVal - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 030	0xE0		0x1E	ManufacturingLocation - SPD read Byte failed

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ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 031	0xE0		0x1F	TrcVal - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 032	0xE0		0x20	TrcdVal - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 033	0xE0		0x21	TaaVal - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 034	0xE0		0x22	CasSup - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 035	0xE0		0x23	SpdIsThermalSensorInstalled - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 036	0xE0		0x24	Available
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 037	0xE0		0x25	DramManufacturerId - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 038	0xE0		0x26	Pmic0Vendor - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 039	0xE0		0x27	Pmic0Revision - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 040	0xE0		0x28	RFC - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 041	0xE0		0x29	IsDdr5SpdCrcValid - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 042	0xE0		0x2A	SPD Manufacturer ID - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 043	0xE0		0x2B	SPD Device Revision Number - SPD read Byte failed

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ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 044	0xE0		0x2C	TSOD Manufacturer ID - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 045	0xE0		0x2D	TSOD Revision Number - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 046	0xE0		0x2E	Get Beta Level - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_READ_MINOR_CODE_ 047	0xE0		0x2F	Get SPD data through HUB - SPD read Byte failed
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_100	0xE0		0x64	Unsupported DeviceType read from SPD
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_101	0xE0		0x65	Unsupported DramDieCount read from SPD
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_102	0xE0		0x66	Unsupported NumColBits read from SPD
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_103	0xE0		0x67	Unsupported NumRowBits read from SPD
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_104	0xE0		0x68	Unsupported Trfc read from SPD
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_105	0xE0		0x69	Unsupported SensorIndex
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_106	0xE0		0x6A	Available
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_107	0xE0		0x6B	Unsupported RawCardRefDesign
ERR_SPD_DECODE/SPD_FATAL _ERROR_UNSUPPORTED_MIN OR_CODE_108	0xE0		0x6C	Unsupported RawCardRefDesignRev

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ERR_SPD_DECODE/SPD_FATAL_ERROR_UNSUPPORTED_MINOR_CODE_109	0xE0		0x6D	Unsupported NominalHeight
ERR_SPD_DECODE/SPD_FATAL_ERROR_MINOR_CODE_MAX	0xE0		0x6E	
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_000	0xE1		0x00	Illegal execution from AP socket in ReleaseExecutionToSocketAp
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_001	0xE1		0x01	Illegal execution from SBSP socket in RespondReadyToSocketSbsp
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_002	0xE1		0x02	Illegal execution from SBSP socket in SendAckToSocketSbsp
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_003	0xE1		0x03	Illegal execution from AP socket in WaitAckFromSocketAp
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_004	0xE1		0x04	Illegal socket execution in GetDdrIoNvDataPointer
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_005	0xE1		0x05	RcAllocatePool failed in SendNvramSocketToAps
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_006	0xE1		0x06	Illegal socket execution in GetDramNvDataPointer
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_007	0xE1		0x07	Available
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_008	0xE1		0x08	Illegal socket execution in GetRcdCwCache
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_009	0xE1		0x09	Illegal socket execution in GetDbCwCache
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_010	0xE1		0x0A	Illegal socket execution in GetFmcCache

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ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_011	0xE1		0x0B	PeiServicesAllocatePages failed to reserve memory for NVRAM in SendPermanentMemoryAddressToAp
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_012	0xE1		0x0C	PeiServicesAllocatePages failed to reserve memory for Training Results in SendPermanentMemoryAddressToAp
ERR_STAGGERED_SYNC/RC_FATAL_ERROR_STAGGERED_SYNC_MINOR_CODE_MAX	0xE1		0x0D	
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_000	0xE2		0x00	RcAllocatePool failed in TrainDcaTco
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_001	0xE2		0x01	RcAllocatePool failed in DcaDfeTrainingMarginDdr5
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_002	0xE2		0x02	Group parameter is not valid when calling SetEyeSizeScalingDfeDdr5Sim
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_003	0xE2		0x03	Group parameter is not valid when calling DfeTrainingDdr5
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_004	0xE2		0x04	GroupInput parameter is not valid when calling EnableDfe
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_005	0xE2		0x05	RcAllocatePool failed in DfeTrainingDdr5
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_006	0xE2		0x06	GroupInput parameter is not valid when calling DfeTrainingDdr5
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_007	0xE2		0x07	Null parameters in DisplayAndAdjustVrefCenteringResultsDdr5
ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_008	0xE2		0x08	RcAllocatePool failed in TrainTcoDqDqs

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_009	0xE2		0x09	Rank out of range in GetMcrPerRankReg
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_010	0xE2		0x0A	Unable to locate gEfiPeiReadOnlyVariable2PpiGuid at LoadImcS3RasData
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_011	0xE2		0x0B	Unable to locate gMemoryPolicyPpiGuid at LoadMemNvram
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_012	0xE2		0x0C	Unexpected LoadUncompressedVariable failure at LoadMemNvram
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_013	0xE2		0x0D	Unable to locate gEfiPeiReadOnlyVariable2PpiGuid at LoadSysInfoNvram
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_014	0xE2		0x0E	Unable to allocating memory at LoadSysInfoNvram
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_015	0xE2		0x0F	Saved SysInfo in Nvram does not match size of struct at LoadSysInfoNvram
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_016	0xE2		0x10	Failed to locate SysInfo pointer from table at LoadSysInfoNvram
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_017	0xE2		0x11	Invalid boot mode at LoadNvramData
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_018	0xE2		0x12	RankList Null pointer at SetCachedRxEq
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_019	0xE2		0x13	RankList Null pointer at GetCachedRxEq
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_020	0xE2		0x14	Null pointer at GetMemoryMode
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_021	0xE2		0x15	Dimm number out of range at DetectLowMargin

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_022	0xE2		0x16	Invalid Channel number at DetectLowMargin
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_023	0xE2		0x17	Host Null pointer at DenormalizeRecvEnable
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_024	0xE2		0x18	ChannelNvList Null pointer at DenormalizeRecvEnable
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_025	0xE2		0x19	DimmNvList Null pointer at DenormalizeRecvEnable
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_026	0xE2		0x1A	Host Null pointer at RenormalizeRecvEnable
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_027	0xE2		0x1B	ChannelNvList Null pointer at RenormalizeRecvEnable
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_028	0xE2		0x1C	DimmNvList Null pointer at RenormalizeRecvEnable
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_029	0xE2		0x1D	unable to locate gMemoryPolicyPpiGuid at HandleAdrEvent
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_030	0xE2		0x1E	RegIndex out of range at GetSetCmdDelay
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_031	0xE2		0x1F	RegIndex out of range at GetSetCmdPieOffset
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_032	0xE2		0x20	ClkMoveMode out of range at DdrtClockMovementWorker
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_033	0xE2		0x21	MaxCmdGroupAll out of range at SetCombinedCmdGroupDutyCycleDdr 5
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_034	0xE2		0x22	Logical Rank out of range at GetSetRxDelayUnicast

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_035	0xE2		0x23	Logical Rank out of range at GetSetRxStrobeInvertUnicast
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_036	0xE2		0x24	Logical Rank out of range at GetSetRxDfeCoefficientTap
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_037	0xE2		0x25	CcSignalsToTrain out of range at GetCcSignalsToTrain
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_038	0xE2		0x26	Invalid input at BuildVrSvidList
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_039	0xE2		0x27	SVID Map is missing an entry at BuildVrSvidList
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_040	0xE2		0x28	SVID Map is built incorrectly at BuildVrSvidList
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_041	0xE2		0x29	VR SVID list length out of range at SetVdd
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_042	0xE2		0x2A	Completion delay out of bounds at SyncCompletionDelayResetCounts
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_043	0xE2		0x2B	Invalid DIMM slot encountered during SetMcDdrCfg
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_044	0xE2		0x2C	Invalid PAGE_TIMER_STATE requested at SetPageTableTimer
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_045	0xE2		0x2D	Null pointer at EnforceDcpmmFreqLimit
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_046	0xE2		0x2E	Null pointer at EnforceDdr5FreqLimit
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_047	0xE2		0x2F	Null pointer at ChipOverrideFreq

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_048	0xE2		0x30	Null pointer at ValidateRasPolicy
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_049	0xE2		0x31	Null pointer at SetPatrolScrub
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_050	0xE2		0x32	Null pointer at DisableClockGatingAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_051	0xE2		0x33	Socket number out of range DisableClockGatingAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_052	0xE2		0x34	Null pointer at DisableAdddcColumnCorrectionAsNe eded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_053	0xE2		0x35	Socket number out of range DisableAdddcColumnCorrectionAsNe eded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_054	0xE2		0x36	Null pointer at SetLinkCfgRead1ReadDataBit18AsNee ded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_055	0xE2		0x37	Socket number out of range SetLinkCfgRead1ReadDataBit18AsNee ded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_056	0xE2		0x38	Null pointer at SetLinkCfgRead1ReaddataBit11AsNee ded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_057	0xE2		0x39	Socket number out of range SetLinkCfgRead1ReaddataBit11AsNee ded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_058	0xE2		0x3A	Null pointer at SetMc0DpChknBit3DisRrIPatsparUcm etaOverwriteAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_059	0xE2		0x3B	Socket number out of range SetMc0DpChknBit3DisRrIPatsparUcm etaOverwriteAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_060	0xE2		0x3C	Null pointer at Wa_14011241978

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_061	0xE2		0x3D	Socket number out of range Wa_14011241978
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_062	0xE2		0x3E	Null pointer at DisableApppErrorLogAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_063	0xE2		0x3F	Socket number out of range DisableApppErrorLogAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_064	0xE2		0x40	Null pointer at DisableWdbParityCheckAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_065	0xE2		0x41	Socket number out of range DisableWdbParityCheckAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_066	0xE2		0x42	Null pointer at DisablePerformanceBasedPreemption AsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_067	0xE2		0x43	Socket number out of range DisablePerformanceBasedPreemption AsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_068	0xE2		0x44	Null pointer at EnablePmon1ForLinkFailAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_069	0xE2		0x45	Socket number out of range EnablePmon1ForLinkFailAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_070	0xE2		0x46	Null pointer at SetMcmainChknBitsDisLclCkGateMcm ainAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_071	0xE2		0x47	Socket number out of range SetMcmainChknBitsDisLclCkGateMcm ainAsNeeded
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_072	0xE2		0x48	Null pointer at SetMc0DpChknBit2DisPersistToSmiFo rDdr
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_073	0xE2		0x49	Socket number out of range SetMc0DpChknBit2DisPersistToSmiFo rDdr

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_074	0xE2		0x4A	Dimm out of range at GetSetThermalRefreshEnables
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_075	0xE2		0x4B	Invalid operation mode at GetSetThermalRefreshEnables
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_076	0xE2		0x4C	Invalid throttle value at GetSetThermalRefreshEnables
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_077	0xE2		0x4D	Dimm out of range at GetSetThermalRefreshThresholds
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_078	0xE2		0x4E	Invalid operation mode at GetSetThermalRefreshThresholds
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_079	0xE2		0x4F	Invalid throttle value at GetSetThermalRefreshThresholds
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_080	0xE2		0x50	Failure read/write CSR at GetSetTurnaroundSettings
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_081	0xE2		0x51	Null pointer at GetSetTurnaroundSettings
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_082	0xE2		0x52	Invalid DIMM slot encountered during SwitchToDdrt2Mode
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_083	0xE2		0x53	Unsupported NvmdimmPerfConfig value at ProgramNvmdimmPerf
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_084	0xE2		0x54	Failure read/write CSR at GetSetDcpmmTurnaroundSettings
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_085	0xE2		0x55	Null pointer at GetSetDcpmmTurnaroundSettings
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_086	0xE2		0x56	MaxRoundTrip out of range at LimitRoundTripAsNeeded

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_087	0xE2		0x57	Null pointer at IsSgxEnabled
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_088	0xE2		0x58	No DDR and HBM memory populated in MemoryPopulationCheck
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_089	0xE2		0x59	Socket number out of range GetRmtResultHobGuid
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_090	0xE2		0x5A	More than one socket or no socket was enabeld in EV test configure
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_091	0xE2		0x5B	IsCadbMcrModeEnabled() called with an invalid parameter
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_092	0xE2		0x5C	CaHwAutoPiSweep() - signal is invalid
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_093	0xE2		0x5D	CaSetHwSweepInit() - signal is invalid
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_094	0xE2		0x5E	BitMargin Null pointer
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_095	0xE2		0x5F	Illegal Max Valid Strobe
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_096	0xE2		0x60	ReadDdrt2WrWrS is not supported
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_097	0xE2		0x61	GetDdrtWriteLatency is not supported
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_098	0xE2		0x62	Invalid condition during MMC pipe communication
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_099	0xE2		0x63	Invalid parameter in RT/IO latency step function

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_100	0xE2		0x64	Unsupported PI ticks per unit in RT/IO latency step function
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_101	0xE2		0x65	Invalid test type in CaClkSetupPiSettingsParams
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_102	0xE2		0x66	MTE IP Init failed
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_103	0xE2		0x67	Unknown MTE pattern
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_104	0xE2		0x68	Invalid mode for setting CADB_BUF
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_105	0xE2		0x69	Invalid signal for setting CADB_BUF CA
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_106	0xE2		0x6A	CachedSelectedId mismatched across subchannels in ProgramSnoopSettingsToDb
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_107	0xE2		0x6B	DDR5_DATA_BUFFER_RW93_REG is tried to be accessed on strobe basis
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_108	0xE2		0x6C	CachedPaged and target page mismatched in WriteDbCwCachePerDimm
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_109	0xE2		0x6D	Invalid MrwAddress parameter in McrDbMr26Mr27Wa()
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_110	0xE2		0x6E	SMBUS Reset during SMBUS read/write
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_111	0xE2		0x6F	IpWrMteGetInst called before IpMteInstData pointer is allocated
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_112	0xE2		0x70	ProgramRddataValidEarly invalid address for DDRINTF_DATA_TIMING1

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ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_113	0xE2		0x71	ProgramRddataValidEarly unsupported rddata_valid_early level
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_114	0xE2		0x72	ProgramSpidRddataValidEarly unsupported rddata_valid_early level
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_115	0xE2		0x73	GetPhysicalMcidLookup invalid parameter
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_116	0xE2		0x74	GetPhysicalMcidLookup unsupported or invalid SOC chop type
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_117	0xE2		0x75	GetPhysicalMcid invalid parameter
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_118	0xE2		0x76	Invalid Mode Only GSM_COMBINE_EDGES mode is expected
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_119	0xE2		0x77	External WL Fine Sweep (No Rising edge found)
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_120	0xE2		0x78	Out of range row failure index in GetRowFailRange
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_121	0xE2		0x79	GetCpgcPrintPatternEnum called with invalid TestType
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_122	0xE2		0x7A	GetCapIdAddressWithErrorHandle() - Invalid input parameter
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_123	0xE2		0x7B	FindBitMappedForGroupRmt() - invalid input Group
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_124	0xE2		0x7C	CpgcMemTestRankSetupMATSRoWRangeWithOffsetCmd() - not 8 BG interleave
ERR_RC_INTERNAL2/RC_FATAL _ERROR_INTERNAL2_MINOR_ CODE_125	0xE2		0x7D	Input BcomVref value of ProgramBcomVref() out of range.

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ERR_RC_INTERNAL2/RC_FATAL_ERROR_INTERNAL2_MINOR_CODE_MAX	0xE2		0x7E	
ERR_RC_DCA_DFE/ERR_INVALID_SIGNAL	0xE6		0x01	
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_000	0xE7		0x00	NULL pointer for ResultProcessing callback
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_001	0xE7		0x01	NULL pointer for BitMargins in PrintCurrentStrobeResults1D
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_002	0xE7		0x02	Channel out of bounds in SkipChannelExecution
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_003	0xE7		0x03	Group delay has been declared as SingleToMultipleDelay and ResultAtRankLevel
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_004	0xE7		0x04	Channel out of bounds in GetDelayPtr
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_005	0xE7		0x05	Channel out of bounds in IsStrobeAtLimit
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_006	0xE7		0x06	Available
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_007	0xE7		0x07	Available
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_008	0xE7		0x08	Available
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_009	0xE7		0x09	Available
ERR_RC_SWEEP_LIB_INTERNAL/RC_FATAL_ERROR_SWEEP_LIB_MINOR_CODE_010	0xE7		0x0A	Illegal value for DeltaOffset in GetVerticalIndex

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ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_011	0xE7		0x0B	No first channel detected
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_012	0xE7		0x0C	NULL pointer for BitMargins in TrackingResults1D
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_013	0xE7		0x0D	NULL pointer for BitMargins in AdjustDitherResults1D
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_014	0xE7		0x0E	NULL pointer for BitMargins in AdjustRelativeResults1D
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_015	0xE7		0x0F	Illegal state in ExecuteStateAction
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_016	0xE7		0x10	Illegal state in GetNextState
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_017	0xE7		0x11	Illegal dither value in ResultsConditionByState
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_018	0xE7		0x12	SubChannelIndex out of bounds in GetStrobeLogical
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_019	0xE7		0x13	StrobeLogical out of bounds in GetStrobeToNibbleOffsetPositionFor Ddr4
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_020	0xE7		0x14	StrobeLogical out of bounds in GetStrobeToNibbleOffsetPositionFor Ddr5
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_021	0xE7		0x15	Per bit value read out during final adjustment not match to original value.
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_022	0xE7		0x16	FLAGS_CATEGORY_RESULTS.ResultsB yPhase is set for a multiknob group delay
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_023	0xE7		0x17	NULL pointer for BitMargins in ConvertResults1D

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ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_024	0xE7		0x18	NULL pointer for BitMargins in ConvertResults1D && AreResultsAtByteLevel
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_025	0xE7		0x19	Invalid logical rank in ClearLegacyResults
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_026	0xE7		0x1A	Invalid ChunkSide in SetStrobeLegacyResult
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_027	0xE7		0x1B	Invalid ChunkSide in SetBitLegacyResult
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_028	0xE7		0x1C	Invalid ChunkSide in ConvertResults1D
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_029	0xE7		0x1D	Invalid ChunkSide in UpdateMaxLeftMinRighEvenOdd
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_030	0xE7		0x1E	Invalid ChunkSide in CleanUpLegacyResults
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_031	0xE7		0x1F	StrobeLogical out of bounds in GetStrobeLogical
ERR_RC_SWEEP_LIB_INTERNAL /RC_FATAL_ERROR_SWEEP_LI B_MINOR_CODE_MAX	0xE7		0x20	
ERR_NO_MEMORY/ERR_NO_ MEMORY_MINOR_NO_MEMO RY	0xE8		0x01	1) No memory was detected via SPD read. No warning log entries available.
				2) invalid config that causes no operable memory. Refer to warning log entries for details.
				MRC Behavior:
				Halts system if there is no memory

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ERR_NO_MEMORY/ERR_NO_MEMORY_MINOR_ALL_CH_DISABLED	0xE8		0x02	Memory on all channels of all sockets are disabled due to hardware memtest error
				MRC Behavior:
				Halts system if there is no memory
ERR_NO_MEMORY/ERR_NO_MEMORY_MINOR_ALL_CH_DISABLED_MIXED	0xE8		0x03	No memory installed. All channels are disabled.
				<i>Not currently used</i>
ERR_LT_LOCK	0xE9	Memory locked due to LT “Bricking”		
		Not currently used		
ERR_DDR_INIT/ERR_RD_DQ_DQS	0xEA		0x01	
ERR_DDR_INIT/ERR_RC_EN	1xEA		0x02	
ERR_DDR_INIT/ERR_WR_LEVEL	2xEA		0x03	
ERR_DDR_INIT/ERR_WR_DQ_DQS	3xEA		0x04	
ERR_DDR_INIT/ERR_TX_RETRAIN_CAP	4xEA		0x05	
ERR_MEM_TEST/ERR_MEM_TEST_MINOR_SOFTWARE	0xEB	Memory test failure	0x01	Software memtest failure.
				<i>Not currently used</i>
ERR_MEM_TEST/ERR_MEM_TEST_MINOR_HARDWARE	0xEB	Memory test failure	0x02	Hardware memtest failed .

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				Not currently used
ERR_MEM_TEST/ERR_MEM_T EST_MINOR_LOCKSTEP_MODE	0xEB	Memory test failure	0x03	Hardware Memtest failure in Lockstep Channel mode requiring a channel
				to be disabled. This is a fatal error which requires a reset and calling MRC
				with a differnt RAS mode to retry.
				Not currently used
ERR_VENDOR_SPECIFIC/ERR_D IMM_PLL_LOCK_ERROR	0xEC	RESERVED for future use	0x01	
ERR_DIMM_COMPAT/ERR_MI XED_MEM_TYPE	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x01	Different dimm types are detected installed in the system
				MRC Behavior:
				System will halt.
ERR_DIMM_COMPAT/ERR_INV ALID_POP	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x02	Violation of population rules
				MRC Behavior:
				System will halt.
ERR_DIMM_COMPAT/ERR_INV ALID_POP_MINOR_QR_AND_3 RD_SLOT	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x03	The 3rd DIMM slot can not be populated when QR DIMMs are installed
				MRC Behavior: System will halt.
ERR_DIMM_COMPAT/ERR_INV	0xED	UDIMMs and RDIMMs are both present DIMM	0x04	UDIMMs and SODIMMs are not supported in the third DIMM slot

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ALID_POP_MINOR_UDIMM_AND_3RD_SLOT		vendor-specific errors		MRC Behavior: System will halt.
ERR_DIMM_COMPAT/ERR_INV_ALID_POP_MINOR_UNSUPPORTED_VOLTAGE	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x05	Unsupported DIMM Voltage
				MRC Behavior:
				System will halt.
ERR_DIMM_COMPAT/ERR_DDR3_DDR4_MIXED	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x06	DDR3 and DDR4 dimms cannot be mixed
				MRC Behavior:
				System will halt.
ERR_DIMM_COMPAT/ERR_MIXED_SPD_TYPE	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x07	Mixing of 256 byte and 512 byte SPD devices is not supported
				MRC Behavior:
				System will halt.
ERR_DIMM_COMPAT/ERR_MISMATCH_DIMM_TYPE	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x08	Mixing of 3DS and non 3DS LRDIMM is not supported
				MRC Behavior:
				System will halt.
ERR_DIMM_COMPAT/ERR_INV_ALID_DDR5_SPD_CONTENT	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x09	
ERR_DIMM_COMPAT/ERR_SM_BUS_READ_FAILURE	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x0A	

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ERR_DIMM_COMPAT/ERR_MIXED_MEM_AEP_AND_UDIMM	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x0B	
ERR_DIMM_COMPAT/ERR_NVMDIMM_NOT_SUPPORTED	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x0C	
ERR_DIMM_COMPAT/ERR_CPU_CAP_NVMDIMM_NOT_SUPPORTED	0xED	UDIMMs and RDIMMs are both present DIMM vendor-specific errors	0x0D	
ERR_MRC_COMPATIBILITY/ERR_MRC_DIR_NONECC	0xEE	Number of HAs found in system greater than MAX_HA defined in MRC build	0x01	RESERVED for future use
		MRC Behavior:		
		System will halt.		
ERR_MRC_STRUCT/ERR_INVALID_BOOT_MODE	0xEF	Indicates a CLTT table structure error. A DIMM	0x01	Boot mode is unknown MRC Behavior: System will halt.
		is populated in the 3rd slot when quad rank DIMM is present in the channel		
		MRC Behavior: System will halt.		
ERR_MRC_STRUCT/ERR_INVALID_SUB_BOOT_MODE	0xEF	Indicates a CLTT table structure error. A DIMM	0x02	MRC Behavior: System will halt. MRC Behavior: System will halt.
		is populated in the 3rd slot when quad rank DIMM is present in the channel		
		MRC Behavior: System will halt.		
ERR_MRC_STRUCT/ERR_INVALID_HOST_ADDR	0xEF	Indicates a CLTT table structure error. A DIMM	0x03	

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		is populated in the 3rd slot when quad rank DIMM is present in the channel		
		MRC Behavior: System will halt.		
ERR_MRC_STRUCT/ ERR_ARRAY_OUT_OF_BOUNDS	0xEF	Indicates a CLTT table structure error. A DIMM	0x04	
		is populated in the 3rd slot when quad rank DIMM is present in the channel		
		MRC Behavior: System will halt.		
ERR_MRC_STRUCT/ERR_IMC_ NUMBER_EXCEEDED	0xEF	Indicates a CLTT table structure error. A DIMM	0x05	
		is populated in the 3rd slot when quad rank DIMM is present in the channel		
		MRC Behavior: System will halt.		
ERR_MRC_STRUCT/ERR_ODT_ STRUCT	0xEF	Indicates a CLTT table structure error. A DIMM	0x06	
		is populated in the 3rd slot when quad rank DIMM is present in the channel		
		MRC Behavior: System will halt.		
ERR_SET_VDD/ERR_UNKNOWN_VR_MODE	0xF0		0x01	Invalid VR mode, unable to set DRAM VDD MRC Behavior: System will halt.
ERR_SET_VDD/ERR_BEYOND_MAX_VDD_OFFSET	0xF0		0x02	

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ERR_IOT_MEM_BUFFER	0xF1	Failure occurred reserving memory for IOT		
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_0	0xF2	Reference code assert	0x00	
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_1	0xF2	Reference code assert	0x01	Failure in InitCpuNvram call
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_2	0xF2	Reference code assert	0x02	Invalid Strobe parameter into GetSetSenseAmpRxOffsetDdrIo
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_3	0xF2	Reference code assert	0x03	Sim register read compare error
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_4	0xF2	Reference code assert	0x04	Sim register read compare error
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_5	0xF2	Reference code assert	0x05	USRA Invalid Register Size
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_6	0xF2	Reference code assert	0x06	Refresh engine is not enabled on S3 exit
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_7	0xF2	Reference code assert	0x07	USRA Invalid Register Size
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_8	0xF2	Reference code assert	0x08	Unsupported function call in MemProjectCommon.c
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_9	0xF2	Reference code assert	0x09	Sim register write compare error
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_10	0xF2	Reference code assert	0x0A	Sim register write compare error
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_11	0xF2	Reference code assert	0x0B	Invalid Bus type for DDRT Sim read
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_12	0xF2	Reference code assert	0x0C	Unable to find register struc for DDRT Sim read
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_13	0xF2	Reference code assert	0x0D	Unable to find register in BDF table for DDRT Sim read

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_14	0xF2	Reference code assert	0x0E	DDRT Sim register read compare error
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_15	0xF2	Reference code assert	0x0F	Invalid Bus type for DDRT Sim write
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_16	0xF2	Reference code assert	0x10	Unable to find register struc for DDRT Sim write
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_17	0xF2	Reference code assert	0x11	Unable to find register in BDF table for DDRT Sim write
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_18	0xF2	Reference code assert	0x12	DDRT Sim register write compare error
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_19	0xF2	Reference code assert	0x13	Unsupported function call in 10nm MemCpgc.c
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_20	0xF2	Reference code assert	0x14	Unsupported function call in 14nm MemCpgc.c
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_21	0xF2	Reference code assert	0x15	Unsupported function call in 14nm CpgclpInterfaces.c
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_22	0xF2	Reference code assert	0x16	Unsupported function call in 10nm MemChipDdrio.c
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_23	0xF2	Reference code assert	0x17	Unsupported function call in 14nm MemChipDdrio.c
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_24	0xF2	Reference code assert	0x18	Unsupported function call in 10nm MemConfigMc.c
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_25	0xF2	Reference code assert	0x19	Unsupported function call in 14nm MemConfigMc.c
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_26	0xF2	Reference code assert	0x1A	Non-SKX CPU installed with SKX build
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_27	0xF2	Reference code assert	0x1B	Unsupported function call in 10nm MemMcChip.c
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_28	0xF2	Reference code assert	0x1C	Get SBSP id for legacy remote SBSP called before Host struc available

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_29	0xF2	Reference code assert	0x1D	AP Socket is trying to access another socket's SMBus
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_30	0xF2	Reference code assert	0x1E	Semaphore call depth rolled over to zero
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_31	0xF2	Reference code assert	0x1F	Invalid BoxType parameter into GetFncCfgAddress
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_32	0xF2	Reference code assert	0x20	Invalid interface parameter into GetFncCfgAddress
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_33	0xF2	Reference code assert	0x21	Invalid interface parameter into ReadFncCfg
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_34	0xF2	Reference code assert	0x22	Invalid interface parameter into WriteFncCfg
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_35	0xF2	Reference code assert	0x23	NULL pointer returned by GetValuesForCurrentConfigOdt()
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_36	0xF2	Reference code assert	0x24	Invalid InterleaveWays value calculated
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_37	0xF2	Reference code assert	0x25	Invalid control group specified
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_38	0xF2	Reference code assert	0x26	Error: Non-production CPU detected. System halted.
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_39	0xF2	Reference code assert	0x27	Error: Unsupported CPU Family and/or Stepping detected. System halted.
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_40	0xF2	Reference code assert	0x28	Invalid IO group parameter into GetCmdGroupAddress
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_41	0xF2	Reference code assert	0x29	Invalid IO group parameter into GetSetCmdDelay
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_42	0xF2	Reference code assert	0x2A	CTL Pi Group not found
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_43	0xF2	Reference code assert	0x2B	Unable to find CMD Pi group in MemCmdControl

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ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_44	0xF2	Reference code assert	0x2C	Unable to find CMD Pi group in MemCmdControl
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_45	0xF2	Reference code assert	0x2D	Attempting to clear TRNG_TARGET with DDRT present in MemCpgc
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_46	0xF2	Reference code assert	0x2E	Attempting to set TRNG_TARGET with no DDRT present in MemCpgc
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_47	0xF2	Reference code assert	0x2F	Invalid training pattern specified in MemCpgc
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_48	0xF2	Reference code assert	0x30	Invalid PatternDepth parameter or NumCacheLines parameter into WriteWDBFixedPatternMATS
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_49	0xF2	Reference code assert	0x31	A17 signal specified for SKX in CpgcCtlClkPattern
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_50	0xF2	Reference code assert	0x32	Accessing NULL pointer in SaveRxPathUngate or RestoreRxPathUngate
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_51	0xF2	Reference code assert	0x33	FindMappedCacheAddressIndex/Writ eTrainingMargin - can not find the mapped cache address index for the input RegAddress/MarginGroup in the FMC/TrainingMarginCache cached list
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_52	0xF2	Reference code assert	0x34	DDR ODT table pointer not found
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_53	0xF2	Reference code assert	0x35	Invalid group parameter into GetSetSenseAmpRxOffsetDdrIo or GetSetDataGroup
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_54	0xF2	Reference code assert	0x36	Invalid group parameter into GetSetTxDelayUnicast
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_55	0xF2	Reference code assert	0x37	Invalid payload parameter into FmvDdrIoInit
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_56	0xF2	Reference code assert	0x38	Invalid bit parameter into GetSetTxDelayBit

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_57	0xF2	Reference code assert	0x39	Insufficient range to handle wrap in MemIOControl
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_58	0xF2	Reference code assert	0x3A	Invalid group parameter into GetSetRxDelayUnicast for CSR read
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_59	0xF2	Reference code assert	0x3B	Invalid group parameter into GetSetRxDelayUnicast for cached read
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_60	0xF2	Reference code assert	0x3C	Invalid group parameter into GetSetRxDelayUnicast for LRDIMM
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_61	0xF2	Reference code assert	0x3D	Invalid bit parameter into GetSetRxDelayBit
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_62	0xF2	Reference code assert	0x3E	Invalid Invalid RTT_WRT ODT setting
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_63	0xF2	Reference code assert	0x3F	Invalid Rtt Wr value stored in Host
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_64	0xF2	Reference code assert	0x40	SimRegsLookup invalid SimRegs group
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_65	0xF2	Reference code assert	0x41	Write to DDR4 LRDIMM with no LRDIMM present
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_66	0xF2	Reference code assert	0x42	Invalid testSize parameter into MemTestMATSN
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_67	0xF2	Reference code assert	0x43	Invalid testSize parameter into MemTestMCH
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_68	0xF2	Reference code assert	0x44	Invalid testSize parameter into MemTestGNDB
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_69	0xF2	Reference code assert	0x45	Invalid testSize parameter into MemTestMARCHCM
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_70	0xF2	Reference code assert	0x46	CTL_CLK_TEST_RANGE is evenly divisible by CTL_CLK_STEPSIZE
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_71	0xF2	Reference code assert	0x47	CLK delay value out of range in GetSetCtlGroupDelayCore

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_72	0xF2	Reference code assert	0x48	CLK delay value out of range in GetSetCmdGroupDelayCore
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_73	0xF2	Reference code assert	0x49	CLK delay value out of range in GetSetClkDelayCore
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_74	0xF2	Reference code assert	0x4A	CLK delay value out of range in GetSetCmdVrefCore
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_75	0xF2	Reference code assert	0x4B	Invalid side parameter in WriteMrsDdr4 with DDRT present
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_76	0xF2	Reference code assert	0x4C	Invalid size parameter in WriteLrbufSmb
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_77	0xF2	Reference code assert	0x4D	Invalid size parameter in WriteLrbufSmb for CTLE
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_78	0xF2	Reference code assert	0x4E	Remote access to another socket's DIMM SPD in multi-threaded mode
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_79	0xF2	Reference code assert	0x4F	Invalid PostCodeMinor in PipeSync
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_80	0xF2	Reference code assert	0x50	Invalid value in Host numRanks in GetCurrentDimmFrequencyAndPopulationOdt (from OdtTableLib)
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_81	0xF2	Reference code assert	0x51	Invalid DIMM specified in GetCurrentDimmFrequencyAndPopulationOdt (from OdtTableLib)
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_82	0xF2	Reference code assert	0x52	Invalid value in Host numRanks in GetActivationForCurrentConfigOdt
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_83	0xF2	Reference code assert	0x53	Invalid DIMM specified in GetActivationForCurrentConfigOdt
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_84	0xF2	Reference code assert	0x54	Invalid Bus Segement specified in ProcSmbChip
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_85	0xF2	Reference code assert	0x55	Invalid SMBUS ClkFrequency Parameter in AdjustProcSmbClkPeriod
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_86	0xF2	Reference code assert	0x56	Free to use

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ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_87	0xF2	Reference code assert	0x57	CacheRoundTrip - Invalid channel specified
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_88	0xF2	Reference code assert	0x58	SetIoLatency - Invalid channel or rank specified
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_89	0xF2	Reference code assert	0x59	GetIoLatency - Invalid channel or rank specified
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_90	0xF2	Reference code assert	0x5A	SetRoundTrip - Invalid channel or rank specified
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_91	0xF2	Reference code assert	0x5B	GetRoundTrip - Invalid channel or rank specified
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_92	0xF2	Reference code assert	0x5C	GetDataTimingRegOffset: Ch or SubCh out of bounds
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_93	0xF2	Reference code assert	0x5D	ProgramQcaDelay: Invalid SubCh
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_94	0xF2	Reference code assert	0x5E	Free to use
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_95	0xF2	Reference code assert	0x5F	Invalid ioGroup parameter into GetCmdGroupAddress10nm
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_96	0xF2	Reference code assert	0x60	GetMCID - invalid parameter
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_97	0xF2	Reference code assert	0x61	SMBUS interface failure in PmicAccessLib
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_98	0xF2	Reference code assert	0x62	New failure occurred in previously masked range in UpdateRowFailures
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_99	0xF2	Reference code assert	0x63	Failure occurred outside of test range in CpgcMemTestMATS
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_100	0xF2	Reference code assert	0x64	ConfigureDramForWriteLeveling - invalid Mode value
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_101	0xF2	Reference code assert	0x65	GetDdrtWriteLatency - Value is less than valid min

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ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_102	0xF2	Reference code assert	0x66	GetDdrtWriteLatency - Value is greater than valid max
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_103	0xF2	Reference code assert	0x67	Error related to use of RC_SIM_FEEDBACK
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_104	0xF2	Reference code assert	0x68	The CWL ADJ delta between the two sub channels is too large and out of range
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_105	0xF2	Reference code assert	0x69	Cannot initialize the DDR5 mode register cache due to an invalid cache pointer
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_106	0xF2	Reference code assert	0x6A	Space allocated for DramNvdata < Space required
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_107	0xF2	Reference code assert	0x6B	Current strobe is out of bounds within subchannel in WriteDramModeRegCachePerDevice
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_108	0xF2	Reference code assert	0x6C	Current strobe is out of bounds within subchannel in ReadDramModeRegCachePerDevice
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_109	0xF2	Reference code assert	0x6D	The Mode Register cache "Valid" array index is out of bounds
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_110	0xF2	Reference code assert	0x6E	The DDR5 mode register is missing from the list of MRs to cache
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_111	0xF2	Reference code assert	0x6F	StrobeLinearToLogical invalid strobe
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_112	0xF2	Reference code assert	0x70	Invalid parameter passed to SetDdrintfCtl0
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_113	0xF2	Reference code assert	0x71	Invalid parameter passed to GetDdrintfCtl0
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_114	0xF2	Reference code assert	0x72	Invalid paramter passed to GetDdrintfCtl0RegAddress
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_115	0xF2	Reference code assert	0x73	Unsupported Sendmrw access
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_116	0xF2	Reference code assert	0x74	CADB2 buffer index out of range

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ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_117	0xF2	Reference code assert	0x75	Trace Hub ACPI base not 256KB aligned
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_118	0xF2	Reference code assert	0x76	Unsupported processor detected in system
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_119	0xF2	Reference code assert	0x77	MemWriteLrdimmBufferHostVrefDq input parameter out of range
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_120	0xF2	Reference code assert	0x78	MemWriteDramVrefDdr5 - invalid input parameter
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_121	0xF2	Reference code assert	0x79	Invalid ch parameter in ProgramJedecTimings
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_122	0xF2	Reference code assert	0x7A	Invalid ch parameter in SetCwlAdj
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_123	0xF2	Reference code assert	0x7B	Invalid Ch parameter in GetIoLatencyIoComp
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_124	0xF2	Reference code assert	0x7C	RestoreRoundTrip - invalid channel specified
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_125	0xF2	Reference code assert	0x7D	Invalid Ch parameter in SetIoLatencyIoComp
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_126	0xF2	Reference code assert	0x7E	ConfigureTxUi - Unable to find RegIndex for signal
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_127	0xF2	Reference code assert	0x7F	SimRegs - Failed to locate register file list instance
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_128	0xF2	Reference code assert	0x80	Memory timing limits not found
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_129	0xF2	Reference code assert	0x81	ConfigureTxUi - Unable to find address for RegIndex
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_130	0xF2	Reference code assert	0x82	Logging power training results failed
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_131	0xF2	Reference code assert	0x83	GetTrainResultStatus - invalid parameter

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ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_132	0xF2	Reference code assert	0x84	Invalid SMBUS controller number
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_133	0xF2	Reference code assert	0x85	SimRegs: Failed to locate socket for PCIE BDF address
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_134	0xF2	Reference code assert	0x86	SimRegs: Failed to locate MemBar entry
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_135	0xF2	Reference code assert	0x87	SimRegs: Memory subsystem type unsupported
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_136	0xF2	Reference code assert	0x88	SimRegs: Failed to locate MMIO region entry
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_137	0xF2	Reference code assert	0x89	SimRegs: Failed to locate register file
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_138	0xF2	Reference code assert	0x8A	SimRegs: Failed to find match for register address
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_139	0xF2	Reference code assert	0x8B	SimRegs: Failed to locate socket for PCIE BDF address
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_140	0xF2	Reference code assert	0x8C	GetSetCmdVrefHostSide called for unknown channel!
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_141	0xF2	Reference code assert	0x8D	Insufficient performance tracker storage
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_142	0xF2	Reference code assert	0x8E	RestoreIoLatency - invalid channel specified
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_143	0xF2	Reference code assert	0x8F	XoverCalib - ProgramOffsets() - Failed to find offset table
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_144	0xF2	Reference code assert	0x90	RTL input is odd and should be even.
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_145	0xF2	Reference code assert	0x91	IO Latency input is even and should be odd.
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_146	0xF2	Reference code assert	0x92	First DIMM not found in an enabled memory channel.

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_147	0xF2	Reference code assert	0x93	Failure to get SMBus address for DIMM slot
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_148	0xF2	Reference code assert	0x94	No valid DDRIO Clock number for NVMDIMM
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_149	0xF2	Reference code assert	0x95	lpmode_entry_latency is set to LPMODE_ENTRY_LATENCY_LARGE_SNR in SNR A0, which is a constraint
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_150	0xF2	Reference code assert	0x96	t_ddrt_cke is set to lpmode_entry_latency-1 in SNR A0, which is a constraint
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_151	0xF2	Reference code assert	0x97	INVALID cmpl_program_delay found
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_152	0xF2	Reference code assert	0x98	InitDdriInterfaceLate - Unreachable code (channel out of bounds)
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_153	0xF2	Reference code assert	0x99	MemCpgc - Row range size is not 2^n
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_154	0xF2	Reference code assert	0x9A	GetSetSxpGnt2Erid - DIMM index is out of bounds
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_155	0xF2	Reference code assert	0x9B	InitializeDefaultData - Invalid number of sockets
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_156	0xF2	Reference code assert	0x9C	DimmInfoDisplayProc - Invalid socket type
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_157	0xF2	Reference code assert	0x9D	GetSetDqRcompOdtDrv/GetSetCccDqRcompOdtDrv - Invalid parameter
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_158	0xF2	Reference code assert	0x9E	InitFnvAccessMode/RestoreFnvAccessMode - Invalid buffer
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_159	0xF2	Reference code assert	0x9F	GetSetTxTco - Invalid level
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_160	0xF2	Reference code assert	0xA0	ProgramMRSAEP - CWL exceeds MR2 bitfield range
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_161	0xF2	Reference code assert	0xA1	Invalid UI delta within the Register Nibble

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ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_162	0xF2	Reference code assert	0xA2	GetPmCmdPwrRegOffset - Index is out of bounds
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_163	0xF2	Reference code assert	0xA3	GetSetRxDelayBit - Writes must specify all bits in the nibble
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_164	0xF2	Reference code assert	0xA4	GetSetTxDelayBit - Writes must specify all bits in the nibble
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_165	0xF2	Reference code assert	0xA5	GetRoundTripRegOffset - Channel out of bounds for DIMM 0
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_166	0xF2	Reference code assert	0xA6	GetRoundTripRegOffset - Channel out of bounds for DIMM 1
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_167	0xF2	Reference code assert	0xA7	GetRoundTripRegOffset - DIMM out of bounds
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_168	0xF2	Reference code assert	0xA8	GetDimmMtrRegOffset - Index out of bounds
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_169	0xF2	Reference code assert	0xA9	GetSetSenseAmpODTDelaysDdr5 - Invalid parameter
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_170	0xF2	Reference code assert	0xAA	MemCpgc Address mode is not supported
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_171	0xF2	Reference code assert	0xAB	QxnCA[m] delay value out of range in GetSetCaBacksideDdr5
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_172	0xF2	Reference code assert	0xAC	QxCS[m] delay value out of range in GetSetCsBacksideDdr5
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_173	0xF2	Reference code assert	0xAD	Size exceeded for RCD SMBUS Write
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_174	0xF2	Reference code assert	0xAE	Current strobe is out of bounds within subchannel in GetStrobeMapping
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_175	0xF2	Reference code assert	0xAF	Current strobe is out of bounds within channel in GetStrobeMapping
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_176	0xF2	Reference code assert	0xB0	Current strobe is out of bounds within channel in GetNibbleFromStrobe

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_177	0xF2	Reference code assert	0xB2	Clock number out of bounds in GetClkRegOffset
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_178	0xF2	Reference code assert	0xB2	ProgramIdleCntValidEarly : Invalid Channel value.
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_179	0xF2	Reference code assert	0xB3	In DCA DFE training, when the count window expires, the CADB is still running.
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_180	0xF2	Reference code assert	0xB4	Refresh mode for TREFI
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_181	0xF2	Reference code assert	0xB5	Invalid parameter into GetSetRxStrobeInvert
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_182	0xF2	Reference code assert	0xB6	DDRT Power cycle is required after frequency changed in CheckDdrtFreqChange
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_183	0xF2	Reference code assert	0xB7	HW RCD DCS Sweep Function not implemented
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_184	0xF2	Reference code assert	0xB8	Invalid Current Sweep Type
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_185	0xF2	Reference code assert	0xB9	JEDEC Arm - Invalid NvDimmEnergyPolicy setup value
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_186	0xF2	Reference code assert	0xBA	EnableDfe - invalid GroupInput parameter
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_187	0xF2	Reference code assert	0xBB	DcaDfelnitDdr5 - invalid Mode parameter
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_188	0xF2	Reference code assert	0xBC	Cannot initialize the DDR5 RCD/DB control word cache or DDRT FMC cache or training margin cache due to an invalid cache pointer
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_189	0xF2	Reference code assert	0xBD	Space allocated for RcdNvdata/DbNvdata/FmcNvData < Space required
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_190	0xF2	Reference code assert	0xBE	The RCD/DB Control Word cache or FMC cache "Valid" array index is out of bounds

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_191	0xF2	Reference code assert	0xBF	RCD/DB CW address not found in the cache list
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_192	0xF2	Reference code assert	0xC0	WARNING - WriteRcdCwCache/WriteDbCwCache called with an invalid parameter
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_193	0xF2	Reference code assert	0xC1	WARNING - ReadRcdCwCache/ReadDbCwCache called with an invalid parameter
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_194	0xF2	Reference code assert	0xC2	EncodedSpdToDramDensity - Invalid input parameter
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_195	0xF2	Reference code assert	0xC3	CompareDimmPartNumberDdr4 - Comparison against the template was incomplete
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_196	0xF2	Reference code assert	0xC4	Invalid group value into EnableTxDfeAllTaps
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_197	0xF2	Reference code assert	0xC5	DDRT Power cycle is needed after freq changed
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_198	0xF2	Reference code assert	0xC6	Invalid signal number
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_199	0xF2	Reference code assert	0xC7	Invalid Group parameter in GetSetRxSamplerEvenOd
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_200	0xF2	Reference code assert	0xC8	Invalid Strobe parameter in GetSetRxSamplerEvenOd
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_201	0xF2	Reference code assert	0xC9	Invaid access for CPGC register cache access
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_202	0xF2	Reference code assert	0xCA	Invalid tCL value got from SPD data for Minimum CAS Latency Time (tAAmin)
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_203	0xF2	Reference code assert	0xCB	Invalid Dimm in the InitAdrForDdrdt
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_204	0xF2	Reference code assert	0xCC	No enabled socket found in the InitAdrTimer
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_205	0xF2	Reference code assert	0xCD	Invid Parameter in DbControlWordWrite/RestoreDbCont

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				rolWordsFromCache
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_206	0xF2	Reference code assert	0xCE	Read Cache before it update by write in AccessDbCwCache
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_207	0xF2	Reference code assert	0xCF	GetDelay - DelayCounts is greater than 32 bits
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_208	0xF2	Reference code assert	0xD0	Invalid Current Test Type
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_209	0xF2	Reference code assert	0xD1	RankListSize out of bounds in SSA CPGC test setup.
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_210	0xF2	Reference code assert	0xD2	DDRIO structure size mismatch
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_211	0xF2	Reference code assert	0xD3	ProgramRxAnalogEnEarly returns non-MRC_STATUS_SUCCESS.
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_212	0xF2	Reference code assert	0xD4	GetIntfDataTiming1RegOffset: Ch or SubCh out of bounds.
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_213	0xF2	Reference code assert	0xD5	Index is out of bounds within subchannel in RcSimCaClkResults or RcSimBacksideCsClkResults
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_214	0xF2	Reference code assert	0xD6	Attempted to release a semaphore that was not owned by the requester
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_215	0xF2	Reference code assert	0xD7	MAX_IMC mismatch with number of MCs reported in CapId3
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_216	0xF2	Reference code assert	0xD8	Invalid GSM_CSN in GetRcdCaSignal
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_217	0xF2	Reference code assert	0xD9	Signal not found in SignalsToSweepDDR5
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_218	0xF2	Reference code assert	0xDA	NULL pointer of NumOfCacheLine in CheckHwSupportedForMATSAndFixupCacheLine for AMT
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_219	0xF2	Reference code assert	0xDB	Available

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_220	0xF2	Reference code assert	0xDC	Signal count more than RCD_CA_SIGNALS_TO_SWEEP
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_221	0xF2	Reference code assert	0xDD	Invalid Dimm, Rank or SubChannel in the GetBacksideCsSignal
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_222	0xF2	Reference code assert	0xDE	Invalid Refresh Mode policy value - DfxDdr5RefreshMode
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_223	0xF2	Reference code assert	0xDF	GetMaxPollingTime - ddrFreq is not in the valid range that can be supported in current
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_224	0xF2	Reference code assert	0xE0	SetCombinedCaGroupDdr5 - unmatched input parameter ValueSize
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_225	0xF2	Reference code assert	0xE1	SetCombinedCmdGroupDdr5 - unmatched input parameter ValueSize
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_226	0xF2	Reference code assert	0xE2	Invalid Dimm, Rank or SubChannel in the GetRcdCsSignal
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_227	0xF2	Reference code assert	0xE3	Illegal strobe in GetSetRxVref
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_228	0xF2	Reference code assert	0xE4	Illegal BitIndex in GetSetRxVref
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_229	0xF2	Reference code assert	0xE5	Invalid LogicalBit in GetDqSwizzles
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_230	0xF2	Reference code assert	0xE6	Invalid TargetStrobe in GetDqSwizzle
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_231	0xF2	Reference code assert	0xE7	Invalid LogicalBit & TargetStrobe combination in GetDqSwizzle
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_232	0xF2	Reference code assert	0xE8	Invalid test type ResultsProcessingDca
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_233	0xF2	Reference code assert	0xE9	Memory allocation failure for RMT core test data
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_234	0xF2	Reference code assert	0xEA	DramModeRegRead run into failure

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ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_235	0xF2	Reference code assert	0xEB	Incorrect format of MRR returned data
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_236	0xF2	Reference code assert	0xEC	GetSetDataBufferDfeTapPerBit - invalid strobe, bit or mode parameter
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_237	0xF2	Reference code assert	0xED	SetDramDutyCycleAdjuster - invalid group parameter
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_238	0xF2	Reference code assert	0xEE	GetSetDramDutyCycleAdjuster - invalid input
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_239	0xF2	Reference code assert	0xEF	GetSetDramDutyCycleAdjuster - invalid strobe or bit parameter
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_240	0xF2	Reference code assert	0xF0	ProcessDramDcaResults - invalid DRAM DCA group
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_241	0xF2	Reference code assert	0xF1	DRAM DCA CalculateEyeWidth - invalid UI/Chunk
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_242	0xF2	Reference code assert	0xF2	Available
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_243	0xF2	Reference code assert	0xF3	GetSetDcaTco - Invalid input parameter SubCh
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_244	0xF2	Reference code assert	0xF4	Invalid parameter Group for GetSetDelayUnicast
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_245	0xF2	Reference code assert	0xF5	Error when GetSetDelayMulticastOffset is executed in GetSetMulticastUnicast as Retry 1
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_246	0xF2	Reference code assert	0xF6	Error while executing unicast function in GetSetDelayUnicast
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_247	0xF2	Reference code assert	0xF7	Error when GetSetDelayMulticastOffset is executed in GetSetMulticastUnicast handling MRC_STATUS_OUT_OF_RESOURCES
ERR_RC_INTERNAL/RC_FATAL_ ERROR_MINOR_CODE_248	0xF2	Reference code assert	0xF8	Error when GetSetDelayMulticastOffset is executed in GetSetRxDelay

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ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_249	0xF2	Reference code assert	0xF9	Error when GetSetDelayMulticastOffset is executed in GetSetRxSamplerEvenOdd
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_250	0xF2	Reference code assert	0xFA	Signal count more than BACKSIDE_CA_SIGNALS_TO_SWEEP
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_251	0xF2	Reference code assert	0xFB	Signal not found in SignalsToSweepBacksideDDR5
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_252	0xF2	Reference code assert	0xFC	Initial CWL Adjust is greater than Preset CWL_ADJ Sweep range
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_253	0xF2	Reference code assert	0xFD	GetParityRecoveryDdr5 not supported for NULL library
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_254	0xF2	Reference code assert	0xFE	GetSetDataBufferDfeTapPerBit - invalid group parameter
ERR_RC_INTERNAL/RC_FATAL_ERROR_MINOR_CODE_255	0xF2	Reference code assert	0xFF	Invalid Rank ID passed to DDRIO Lib function
ERR_RC_INTERNAL/ERR_RC_INTERNAL_HBM	0xF2	Reference code assert	0x01	
ERR_INVALID_REG_ACCESS/ERR_INVALID_WRITE_REG_BDF	0xF3		0x01	Unsupported MC frequency set MRC Behavior: System will halt.
ERR_INVALID_REG_ACCESS/ERR_INVALID_WRITE_REG_BDF	0xF3		0x02	
ERR_INVALID_REG_ACCESS/ERR_INVALID_WRITE_REG_OFFSET	0xF3		0x03	
ERR_INVALID_REG_ACCESS/ERR_INVALID_READ_REG_BDF	0xF3		0x04	
ERR_INVALID_REG_ACCESS/ERR_INVALID_READ_REG_OFFSET	0xF3		0x05	
ERR_INVALID_REG_ACCESS/ERR_INVALID_WRITE_REG_SIZE	0xF3		0x06	
ERR_INVALID_REG_ACCESS/ERR_INVALID_READ_REG_SIZE	0xF3		0x07	

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ERR_INVALID_REG_ACCESS/ERR_UNKNOWN_REG_TYPE	0xF3		0x08	
ERR_INVALID_REG_ACCESS/ERR_INVALID_ACCESS_METHOD	0xF3		0x09	
ERR_SET_MC_FREQ/ERR_UNSUPPORTED_MC_FREQ	0xF4		0x01	Unable to get current MC frequency MRC Behavior: System will halt.
ERR_SET_MC_FREQ/ERR_UNSPECIFIED_MC_FREQ_SETTING_ERROR	0xF4		0x02	
ERR_READ_MC_FREQ/ERR_NOTABLE_READ_MC_FREQ	0xF5		0x01	
ERR_DIMM_CHANNEL	0x70			
ERR_BIST_CHECK	0x74			
ERR_DDR_FREQ_NOT_FOUND	0x75			
ERR_PIPE	0x76			
ERR_SMBUS/TSOD_POLLING_ENABLED_READ	0xF6		0x01	
ERR_SMBUS/TSOD_POLLING_ENABLED_WRITE	0xF6		0x02	
ERR_SMBUS/ERR_LRDIMM_SMBUS_READ_FAILURE	0xF6		0x03	
ERR_PCU/PCU_NOT_RESPONDING	0xF7		0x01	
ERR_PCU/FUSE_ERROR	0xF7		0x02	
ERR_PCU/ERR_PCU_COMMAND_NOT_SUPPORTED	0xF7		0x03	

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ERR_NGN/NGN_DRIVER_NOT_RESPONSIBLE	0xF8		0x01	
ERR_NGN/NGN_ARRAY_OUT_OF_BOUNDS	0xF8		0x02	
ERR_NGN/NGN_PMEM_CONFIG_ERROR	0xF8		0x03	
ERR_NGN/INTERLEAVE_EXCEEDED	0xF8		0x04	
ERR_NGN/NGN_BYTES_MISMATCH	0xF8		0x05	
ERR_NGN/NGN_SKU_MISMATCH	0xF8		0x06	
ERR_INTERLEAVE_FAILURE/ERR_RIR_RULES_EXCEEDED	0xF9		0x01	
ERR_CPGC_TIMEOUT	0xFA			
ERR_CAR_LIMIT/ERR_OUT_OF_CAR_RESOURCES	0xFB		0x01	
PRINTF_OUTOF_SYNC_ERROR_MAJOR/PRINTF_OUTOF_SYNC_ERROR_MINOR	0xCF		0x01	
ERR_CMI_FAILURE/ERR_CMI_INIT_FAILED	0xFC		0x01	
ERR_VALUE_OUT_OF_RANGE/ERR_VALUE_BELOW_MIN	0xFD		0x01	
ERR_VALUE_OUT_OF_RANGE/ERR_VALUE_ABOVE_MAX	0xFD		0x02	
ERR_DDRIO_HWFSM/ERR_XOVER_HWFSM_TIMEOUT	0xFE		0x01	
ERR_DDRIO_HWFSMERR_XOVER_HWFSM_FAILURE	0xFE		0x02	

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ERR_DDRIO_HWFSMERR_SENS EAMP_HWFSM_TIMEOUT	0xFE		0x03	
ERR_DDRIO_HWFSMERR_SENS EAMP_HWFSM_FAILURE	0xFE		0x04	
ERR_DDRIO_HWFSMERR_SENS EAMP_HWFSM_SIGNAL_ERRO R	0xFE		0x05	
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_000	0xFF		0x00	CheckValueForPda - invalid BestTapBias pointer
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_001	0xFF		0x01	ProgramDfeTapBias - invalid BestTapBias pointer instance 1
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_002	0xFF		0x02	ProgramDfeTapBias - invalid BestTapBias pointer instance 2
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_003	0xFF		0x03	GetSetProtectedDcpmmRegister - Value pointer invalid
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_004	0xFF		0x04	ConvertValueToDramDutyCycleAdjust erRegFormat - bad pointer
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_005	0xFF		0x05	WriteProtectedDcpmmRegister - bad pointer
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_006	0xFF		0x06	ProgramReadDfeTapBias - bad BestTapBias pointer instance 1
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_007	0xFF		0x07	ProgramReadDfeTapBias - bad BestTapBias pointer instance 2
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_008	0xFF		0x08	RcAllocatePool failed in DcsDcaQcsDdriolnitConfiguration
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_009	0xFF		0x09	GetDdr5LrdimmTrainingData failed

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ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_010	0xFF		0x0A	GetCsRange - TestRange null pointer
ERR_MRC_POINTER/RC_FATAL _ERROR_MRC_POINTER_MINO R_CODE_MAX	0xFF		0x0B	
Memory Warn Code				
WARN_RDIMM_ON_UDIMM/ WARN_MINOR_CONFIG_NOT_ SUPPORTED	0x01		0x01	
WARN_UDIMM_ON_RDIMM/ WARN_MINOR_CONFIG_NOT_ SUPPORTED	0x02		0x01	
WARN_SODIMM_ON_RDIMM/ WARN_MINOR_CONFIG_NOT_ SUPPORTED	0x03		0x01	
WARN_4Gb_FUSE	0x04			
WARN_8Gb_FUSE	0x05			
WARN_IMC_DISABLED	0x06			
WARN_DIMM_COMPAT/WAR N_DIMM_COMPAT_MINOR_X 16_COMBO	0x07		0x01	
WARN_DIMM_COMPAT/WAR N_DIMM_COMPAT_MINOR_M AX_RANKS	0x07		0x02	
WARN_DIMM_COMPAT/WAR N_DIMM_COMPAT_MINOR_Q R	0x07		0x03	
WARN_DIMM_COMPAT/WAR N_DIMM_COMPAT_MINOR_N OT_SUPPORTED	0x07		0x04	
WARN_DIMM_COMPAT/WAR N_RANK_NUM	0x07		0x05	

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WARN_DIMM_COMPAT/WARN_TOO_SLOW	0x07		0x06	
WARN_DIMM_COMPAT/WARN_DIMM_COMPAT_MINOR_ROWS_ADDR_ORDER	0x07		0x07	
WARN_DIMM_COMPAT/WARN_CHANNEL_CONFIG_NOT_SUPPORTED	0x07		0x08	
WARN_DIMM_COMPAT/WARN_CHANNEL_MIX_ECC_NONECC	0x07		0x09	
WARN_DIMM_COMPAT/WARN_DIMM_VOLTAGE_NOT_SUPPORTED	0x07		0x0A	
WARN_DIMM_COMPAT/WARN_DIMM_COMPAT_TRP_NOT_SUPPORTED	0x07		0x0B	
WARN_DIMM_COMPAT/WARN_DIMM_NONECC	0x07		0x0C	
WARN_DIMM_COMPAT/WARN_DIMM_COMPAT_3DS_RDIMM_NOT_SUPPORTED	0x07		0x0D	
WARN_DIMM_COMPAT/WARN_RANK_COUNT_MISMATCH	0x07		0x0E	
WARN_DIMM_COMPAT/WARN_DIMM_SKU_MISMATCH	0x07		0x0F	
WARN_DIMM_COMPAT/WARN_DIMM_3DS_NOT_SUPPORTED	0x07		0x10	
WARN_DIMM_COMPAT/WARN_DIMM_NVMDIMM_NOT_SUPPORTED	0x07		0x11	
WARN_DIMM_COMPAT/WARN_DIMM_16GB_SUPPORTED	0x07		0x12	
WARN_DIMM_COMPAT/WARN_CHANNEL_SKU_NOT_SUPPORTED	0x07		0x13	

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WARN_DIMM_COMPAT/WARN_DIMM_SKU_NOT_SUPPORTED	0x07		0x14	
WARN_DIMM_COMPAT/WARN_CHANNEL_CONFIG_FREQ_NOT_SUPPORTED	0x07		0x15	
WARN_DIMM_COMPAT/WARN_DIMM_SPEED_NOT_SUPPORTED	0x07		0x16	
WARN_DIMM_COMPAT/WARN_NO_DDR4_ON_SOCKET	0x07		0x17	
WARN_DIMM_COMPAT/WARN_NO_DDR4_ON_SOCKET0	0x07		0x18	
WARN_DIMM_COMPAT/WARN_DIMM_NOT_IN_DDRT_PORT_DDR_TABLE	0x07		0x19	
WARN_DIMM_COMPAT/WARN_SOCKET_CPU_DOES_NOT_SUPPORT_NVMDIMM	0x07		0x1A	
WARN_DIMM_COMPAT/WARN_SOCKET_SKU_DOES_NOT_SUPPORT_NVMDIMM	0x07		0x1B	
WARN_DIMM_COMPAT/WARN_DIMM_IN_DDRT_PORT_DDR_TABLE_NOT_VALIDATED	0x07		0x1C	
WARN_DIMM_COMPAT/WARN_DDRT_DIMM_NOT_SUPPORTED	0x07		0x1D	
WARN_DIMM_COMPAT/WARN_CHANNEL_WIDTH_MIX_NOT_SUPPORTED	0x07		0x1E	
WARN_DIMM_COMPAT/WARN_DIMM_24GB_SUPPORTED	0x07		0x1F	
WARN_DIMM_COMPAT/WARN_DIMM_32GB_SUPPORTED	0x07		0x20	
WARN_DIMM_COMPAT/WARN_LRDIMM_NOT_SUPPORTED	0x07		0x21	

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WARN_LOCKSTEP_DISABLE/WARN_LOCKSTEP_DISABLE_MINOR_RAS_MODE	0x09		0x01	
WARN_LOCKSTEP_DISABLE/WARN_LOCKSTEP_DISABLE_MINOR_MISMATCHED	0x09		0x02	
WARN_LOCKSTEP_DISABLE/WARN_LOCKSTEP_DISABLE_MINOR_MEMTEST_FAILED	0x09		0x03	
WARN_USER_DIMM_DISABLE/WARN_USER_DIMM_DISABLE_QUAD_AND_3DPC	0x0A		0x01	
WARN_USER_DIMM_DISABLE/WARN_USER_DIMM_DISABLE_MEMTEST	0x0A		0x02	
WARN_USER_DIMM_DISABLE/WARN_USER_DIMM_DISABLE_RANK_DISABLED	0x0A		0x03	
WARN_USER_DIMM_DISABLE/WARN_USER_DIMM_DISABLE_MAPPED_OUT	0x0A		0x04	
WARN_MEMTEST_DIMM_DISABLE	0x0B			
WARN_MIRROR_DISABLE/WARN_MIRROR_DISABLE_MINOR_RAS_DISABLED	0x0C		0x01	
WARN_MIRROR_DISABLE/WARN_MIRROR_DISABLE_MINOR_MISMATCH	0x0C		0x02	
WARN_MIRROR_DISABLE/WARN_MIRROR_DISABLE_MINOR_MEMTEST	0x0C		0x03	
WARN_PMIRROR_DISABLE/WARN_MIRROR_DISABLE_MINOR_RAS_DISABLED	0x0D		0x01	
WARN_INTERLEAVE_FAILURE/WARN_SAD_RULES_EXCEEDED	0x0e		0x01	

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WARN_INTERLEAVE_FAILURE/ WARN_TAD_RULES_EXCEEDED	0x0e		0x02	
WARN_INTERLEAVE_FAILURE/ WARN_RIR_RULES_EXCEEDED	0x0e		0x03	
WARN_INTERLEAVE_FAILURE/ WARN_TAD_OFFSET_NEGATIVE	0x0e		0x04	
WARN_INTERLEAVE_FAILURE/ WARN_TAD_LIMIT_ERROR	0x0e		0x05	
WARN_INTERLEAVE_FAILURE/ WARN_INTERLEAVE_3WAY	0x0e		0x06	
WARN_INTERLEAVE_FAILURE/ WARN_A7_MODE_AND_3WAY _CH_INTRLV	0x0e		0x07	
WARN_INTERLEAVE_FAILURE/ WARN_INTERLEAVE_EXCEEDED	0x0e		0x08	
WARN_INTERLEAVE_FAILURE/ WARN_DIMM_CAPACITY_MISMATCH	0x0e		0x09	
WARN_INTERLEAVE_FAILURE/ WARN_DIMM_POPULATION_MISMATCH	0x0e		0x0A	
WARN_INTERLEAVE_FAILURE/ WARN_NM_MAX_SIZE_EXCEEDED	0x0e		0x0B	
WARN_INTERLEAVE_FAILURE/ WARN_NM_SIZE_BELOW_MIN_LIMIT	0x0e		0x0C	
WARN_INTERLEAVE_FAILURE/ WARN_NM_SIZE_NOT_POWER_OF_TWO	0x0e		0x0D	
WARN_INTERLEAVE_FAILURE/ WARN_MAX_INTERLEAVE_SETS_EXCEEDED	0x0e		0x0E	
WARN_INTERLEAVE_FAILURE/ WARN_NGN_DIMM_COMM_FAILED	0x0e		0x0F	

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WARN_DIMM_COMM_FAILED /WARN_MINOR_DIMM_COM M_START_DOORBELL_TIMEOU T	0x0F		0x01	
WARN_DIMM_COMM_FAILED /WARN_MINOR_DIMM_COM M_FAILED_STATUS	0x0F		0x02	
WARN_DIMM_COMM_FAILED /WARN_MINOR_DIMM_MAILB OX_FAILED	0x0F		0x03	
WARN_DIMM_COMM_FAILED /WARN_MINOR_DIMM_COM M_FINISH_DOORBELL_TIMEOU T	0x0F		0x04	
WARN_DIMM_COMM_FAILED /WARN_MINOR_DIMM_COM M_FINISH_COMPLETE_TIMEO UT	0x0F		0x05	
WARN_SPARE_DISABLE/WARN _SPARE_DISABLE_MINOR_RK_ SPARE	0x10		0x01	
WARN_PTRLSCRB_DISABLE/W ARN_PTRLSCRB_MINOR_DISAB LE	0x11		0x00	
WARN_UNUSED_MEMORY/W ARN_UNUSED_MEMORY_MIN OR_MIRROR	0x12		0x01	
WARN_UNUSED_MEMORY/W ARN_UNUSED_MEMORY_MIN OR_LOCKSTEP	0x12		0x02	
WARN_RD_DQ_DQS	0x13			
WARN_RD_RCVEN/WARN_RO UNDTRIP_EXCEEDED	0x14		0x01	
WARN_RD_RCVEN/WARN_RCV EN_PI_DELAY_EXCEEDED	0x14		0x02	
WARN_WR_LEVEL/WARN_WR _FLYBY_CORR	0x15		0x00	

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WARN_WR_LEVEL/WARN_WR_FLYBY_UNCORR	0x15		0x01	
WARN_WR_LEVEL/WARN_WR_FLYBY_DELAY	0x15		0x02	
WARN_WR_DQ_DQS	0x16			
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_MINOR_OUT_OF_ORDER	0x17		0x01	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_MINOR_INDEPENDENT_MODE	0x17		0x02	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_2_AEP_FOUNDED_ON_SAME_CH	0x17		0x03	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_MINOR_MIXED_RANKS_FOUND	0x17		0x04	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_NVMDIMM_OUT_OF_ORDER	0x17		0x20	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_UDIMM_POPULATION	0x17		0x40	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_RDIMM_POPULATION	0x17		0x80	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_LRDIMM_DUAL_DIE_POPULATION	0x17		0x05	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_MINOR_NO_DDRT_ON_MC0	0x17		0x06	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_MINOR_NO_DDRT_ON_SOCKET0	0x17		0x07	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_DDR4_CA	0x17		0x08	

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P_NOT_POR_FOR_MEM_POP				
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_DDR4_TYPE_NOT_POR_FOR_MM	0x17		0x09	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_MEM_TOPOLOGY_NOT_SYMMETRICAL	0x17		0x0A	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_NM_FM_RATIO_VIOLATION	0x17		0x0B	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_2LM_IMC_MEM_MISMATCH	0x17		0x0C	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_DDRT_CAPACITY_MISMATCH	0x17		0x0D	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_DDRT_PARTITION_MISMATCH	0x17		0x0E	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_AEP_VDD_CHANGED	0x17		0x0F	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_SOCKET_MODE_MISMATCH	0x17		0x10	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_DCPDIMM_MIXING	0x17		0x11	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_DDR_CAPACITY_MISMATCH	0x17		0x12	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_POPULATION_POR_MISMATCH	0x17		0x13	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_MEM_POP_POR_TBL_INVALID	0x17		0x14	

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WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_2LM_FM_CH_NOT_PWR_OF_TWO	0x17		0x15	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_POPULATION_POR_NOT_FOUND	0x17		0x16	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_PMEM_X1_POPULATION_INVALID	0x17		0x17	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_NO_DIMM_ON_SOCKET	0x17		0x18	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_UMA_POWER_COMPATABILITY	0x17		0x19	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_MCR_POPULATION_INVALID	0x17		0x1A	
WARN_DIMM_POP_RUL/WARN_DIMM_POP_RUL_9X4_CONFIG_INVALID	0x17		0x1B	
WARN_DIMM_POP_RUL_EXTENDED/WARN_DIMM_POP_RUL_LRDIMM_3DS_POPULATION	0x3E		0x02	
WARN_DIMM_POP_RUL_EXTENDED/WARN_DIMM_POP_RUL_RDIMM_3DS_POPULATION	0x3E		0x04	
WARN_CLTT_DISABLE/WARN_CLTT_MINOR_NO_TEMP_SENSOR	0x18		0x01	
WARN_CLTT_DISABLE/WARN_CLTT_MINOR_NO_TEMP_SENSOR	0x18		0x02	
WARN_THROT_INSUFFICIENT/WARN_2X_REFRESH_TEMPLO_DISABLED	0x19		0x01	
WARN_THROT_INSUFFICIENT/WARN_CUSTOM_REFRESH_RATE_REVERTED	0x19		0x02	

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WARN_CLTT_DIMM_UNKNOWN	0x1A			
WARN_DQS_TEST	0x1B			
WARN_MEM_TEST	0x1C			
WARN_CLOSED_PAGE_OVERRIDE	0x1D			
WARN_DIMM_VREF_NOT_PRESSENT	0x1E			
WARN_EARLY_RID/WARN_EARLY_RID_UNCORR	0x1F		0x01	
WARN_EARLY_RID/WARN_EARLY_RID_CYCLE_FAIL	0x1F		0x02	
WARN_LV_STD_DIMM_MIX	0x20			
WARN_LV_2QR_DIMM	0x21			
WARN_LV_3DPC	0x22			
WARN_CMD_ADDR_PARITY_ERROR	0x23			
WARN_DQ_SWIZZLE_DISC/WARN_DQ_SWIZZLE_DISC_UNCORR	0x24		0x01	
WARN_COD_HA_NOT_ACTIVE	0x25			
WARN_CMD_CLK_TRAINING/WARN_CMD_PI_GROUP_SMALL_EYE	0x26		0x01	
WARN_CMD_CLK_TRAINING/WARN_CMD_PI_GROUP_NO_EYE	0x26		0x02	

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WARN_INVALID_BUS	0x27			
WARN_INVALID_FNV_OPCODE /WARN_INVALID_FNV_SUBOP CODE	0x28		0x00	
WARN_MEMORY_TRAINING/ WARN_CTL_CLK_LOOPBACK_T RAINING	0x29		0x02	
WARN_MEMORY_TRAINING/ WARN_ODT_TIMING_OVERFL OW	0x29		0x03	
WARN_MEMORY_TRAINING/ WARN_CS_CLK_LOOPBACK_TR AINING	0x29		0x04	
WARN_MEMORY_TRAINING/ WARN_CA_CLK_LOOPBACK_TR AINING	0x29		0x05	
WARN_NO_DDR_MEMORY/W ARN_NO_MEMORY_MINOR_N O_MEMORY	0x2A		0x01	
WARN_NO_DDR_MEMORY/W ARN_NO_MEMORY_MINOR_A LL_CH_DISABLED	0x2A		0x02	
WARN_NO_DDR_MEMORY/W ARN_NO_MEMORY_MINOR_A LL_CH_DISABLED_MIXED	0x2A		0x03	
WARN_NO_DDR_MEMORY/W ARN_NO_MEMORY_MINOR_N O_DDR	0x2A		0x04	
WARN_ROUNDTRIP_ERROR/W ARN_RCVNTAP_CMDDELAY_EX CEEDED	0x2B		0x01	
WARN_MEMORY_MODEL_ERR OR/WARN_SNC24_MODEL_ER ROR	0x2C		0x01	
WARN_MEMORY_MODEL_ERR OR/WARN_QUAD_HEMI_MOD EL_ERROR	0x2C		0x02	

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WARN_MEMORY_MODEL_ERROR/WARN_SNC24_DIMM_POPULATION_MISMATCH	0x2C		0x03	
WARN_MEMORY_MODEL_ERROR/WARN_SNC24_INCOMPATIBLE_DDR_CAPACITY	0x2C		0x04	
WARN_MEMORY_MODEL_ERROR/WARN_SNC24_INCOMPATIBLE_MCDRAM_CAPACITY	0x2C		0x05	
WARN_MEMORY_MODEL_ERROR/WARN_MCDRAM_CONFIG_NOT_SUPPORTED	0x2C		0x06	
WARN_MEMORY_MODEL_ERROR/WARN_SNC24_TILE_POPULATION_MISMATCH	0x2C		0x07	
WARN_OVERRIDE_MEMORY_MODE/WARN_OVERRIDE_TO_FLAT_NOT_ENOUGH_NEAR_MEMORY	0x2D		0x01	
WARN_OVERRIDE_MEMORY_MODE/WARN_OVERRIDE_TO_FLAT_NOT_ENOUGH_FAR_MEMORY	0x2D		0x02	
WARN_OVERRIDE_MEMORY_MODE/WARN_OVERRIDE_TO_HYBRID_50_PERCENT	0x2D		0x03	
WARN_OVERRIDE_MEMORY_MODE/WARN_OVERRIDE_TO_FLAT_NO_MCDRAM_MEMORY	0x2D		0x01	
WARN_OVERRIDE_MEMORY_MODE/WARN_OVERRIDE_TO_FLAT_NO_DDR_MEMORY	0x2D		0x02	
WARN_OVERRIDE_MEMORY_MODE/WARN_OVERRIDE_TO_HYBRID_CACHESIZE_1_2	0x2D		0x03	
WARN_MEM_INIT	0x2E			
WARN_SENS_AMP_TRAINING/WARN_SENS_AMP_CH_FAILIU	0x2F		0x01	

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RE				
WARN_FPT_CORRECTABLE_ERROR	0x30			
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_R D_DQ_DQS	0x31		0x13	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_R D_RCVEN	0x31		0x14	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_W R_LEVEL	0x31		0x15	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_W R_FLYBY	0x31		0x00	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_W R_DQ_DQS	0x31		0x16	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_D QS_TEST	0x31		0x1B	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_M EM_TEST	0x31		0x1C	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_DWL_EXT_COARSE_TRAINING	0x31		0x20	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_DWL_EXT_FINE_TRAINING	0x31		0x21	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_DWL_INT_COARSE_TRAINING	0x31		0x22	

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WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_DWL_INT_FINE_TRAINING	0x31		0x23	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_TRAINING	0x31		0x24	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_V REF_TRAINING	0x31		0x25	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_RCVEN_PHASE_TRAINING	0x31		0x26	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_RCVEN_CYCLE_TRAINING	0x31		0x27	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_READ_DELAY_TRAINING	0x31		0x28	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_WL_TRAINING	0x31		0x29	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_COARSE_WL_TRAINING	0x31		0x2A	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_LR DIMM_WRITE_DELAY_TRAINING	0x31		0x2B	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_QxCA_CLK_NO _EYE_FOUND	0x31		0x2C	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_ROW_FAILURE	0x31		0x2D	

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WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_PPR_ROW_REPAIR	0x31		0x2E	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_WL_EXTERNAL_COARSE	0x31		0x30	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_WL_EXTERNAL_FINE	0x31		0x31	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_WL_INTERNAL_COARSE	0x31		0x32	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_WL_INTERNAL_FINE	0x31		0x33	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_WL_INTERNAL_OUT_OF_CYCLE	0x31		0x34	
WARN_FPT_UNCORRECTABLE_ERROR/WARN_FPT_MINOR_RD_DQ_DQS_JITTER	0x31		0x35	
WARN_CH_DISABLED/WARN_TWR_LIMIT_REACHED	0x32		0x01	
WARN_CH_DISABLED/WARN_TWR_LIMIT_ON_LOCKSTEP_CH	0x32		0x02	
WARN_CH_DISABLED/WARN_SPD_BLOCK_UNLOCKED	0x32		0x03	
WARN_MEM_LIMIT	0x33			
WARN_RT_DIFF_EXCEED/WARN_RT_DIFF_MINOR_EXCEED	0x34		0x01	
WARN_PPR_FAILED	0x35			
WARN_REGISTER_OVERFLOW/WARN_MINOR_REGISTER_OVERFLOW	0x36		0x01	

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WARN_REGISTER_OVERFLOW/ WARN_MINOR_REGISTER_UN DERFLOW	0x36		0x02	
WARN_SWIZZLE_DISCOVERY_T RAINING/WARN_SWIZZLE_PAT TERN_MISMATCH	0x37		0x01	
WARN_WRCRC_DISABLE/WAR N_TRAIL_ODT_LIMIT_REACHE D	0x38		0x01	
WARN_FNV_BSR/WARN_DT_E RROR	0x39		0x01	
WARN_FNV_BSR/WARN_MEDI A_READY_ERROR	0x39		0x02	
WARN_FNV_BSR/WARN_POLLI NG_LOOP_TIMEOUT	0x39		0x03	
WARN_FNV_BSR/WARN_OPCO DE_INDEX_LOOKUP	0x39		0x04	
WARN_FNV_BSR/WARN_DR_R EADY_ERROR	0x39		0x05	
WARN_ADDDC_DISABLE/WAR N_ADDDC_MINOR_DISABLE	0x3A		0x01	
WARN_SDDC_DISABLE/WARN_ SDDC_MINOR_DISABLE	0x3B		0x02	
WARN_FW_OUT_OF_DATE/W ARN_FW_CLK_MOVEMENT	0x3C		0x00	
WARN_FW_OUT_OF_DATE/W ARN_FW_BCOM_MARGINING	0x3C		0x01	
WARN_SMBUS_FAULT/WARN_ SMBUS_WRITE	0x3D		0x01	
WARN_SMBUS_FAULT/WARN_ SMBUS_READ	0x3D		0x02	
WARN_COMPLETION_DELAY_E RROR/WARN_CMPL_DELAY_BE LOW_MIN	0x3F		0x02	

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WARN_COMPLETION_DELAY_ERROR/WARN_CMPL_DELAY_MAX_EXCEEDED	0x3F		0x03	
WARN_MEM_CONFIG_CHANGED/WARN_MEM_OVERRIDE_DISABLED	0x40		0x01	
WARN_DIMM_MEM_MODE/WARN_DIMM_MEM_MODE_NEW_DIMM_2LM_NOT_SUPPORTED	0x41		0x01	
WARN_DIMM_MEM_MODE/WARN_DIMM_MEM_MODE_2LM_NOT_SUPPORTED	0x41		0x02	
WARN_DFE_TAP_MARGIN_STATUS/WARN_DFE_TAP_ZERO_MARGIN_ERROR	0x42		0x01	
WARN_DFE_MARGIN_STATUS/WARN_DFE_ZERO_MARGIN_ERROR	0x43		0x01	
WARN_FREQUENCY_POR_CHECK/WARN_PLATFORM_SEGMENT_NOT_FOUND	0x44		0x00	
WARN_FREQUENCY_POR_CHECK/WARN_INVALID_NUMBER_OF_SOCKETS	0x44		0x01	
WARN_FREQUENCY_POR_CHECK/WARN_UNKNOWN_PLATFORM_SEGMENT	0x44		0x02	
WARN_FREQUENCY_POR_CHECK/WARN_INVALID_POR_SOCKET_CONFIG	0x44		0x03	
WARN_BOUNDS_ERROR/WARN_INVALID_DIMM_INDEX	0x45		0x01	
WARN_POWER_FAILURE/WARN_DDR5_POWER_FAILURE_PHASE_1	0x46		0x00	
WARN_POWER_FAILURE/WARN_DDR5_POWER_FAILURE_PHASE_2	0x46		0x01	

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WARN_POWER_FAILURE/WARN_DDRT_POWER_FAILURE	0x46		0x02	
WARN_POWER_FAILURE/WARN_DDRIO_POWER_FAILURE	0x46		0x03	
WARN_TRAINING_RESULT_CHECK/WARN_TRAINING_RESULT_BEYOND_LIMIT	0x47		0x01	
WARN_MEMTEST_ROW_FAIL_RANGE_ERROR/WARN_MEMTEST_DDR_ROW_FAIL_RANGE_LIST_FULL	0x48		0x01	
WARN_MEMTEST_ROW_FAIL_RANGE_ERROR/WARN_MEMTEST_HBM_ROW_FAIL_RANGE_LIST_FULL	0x48		0x02	
WARN_MCA_UNCORRECTABLE_ERROR/WARN_PREVIOUS_BOOT_MCA_MINOR_CODE	0x50		0x01	
WARN_DM_TEST_ERROR_CODE/WARN_DM_TEST_PARSE_ERROR_MINOR_CODE	0x60		0x01	
WARN_DM_TEST_ERROR_CODE/WARN_DM_TEST_CONFIGURATION_ERROR_MINOR_CODE	0x60		0x02	
WARN_DM_TEST_ERROR_CODE/WARN_DM_TEST_EXECUTION_ERROR_MINOR_CODE	0x60		0x03	
WARN_FPGA_NOT_DETECTED	0x75			
WARN_HBM_POR_LIMITATION/WARN_HBM_UNSUPPORTED_SOCKET_POPULATION	0x79		0x01	
WARN_HBM_POR_LIMITATION/WARN_HBM_UNSUPPORTED_IO_STACK_POPULATION	0x79		0x02	
WARN_FSM_ERROR_CODE/WARN_FSM_TIMEOUT_ERROR	0x80		0x01	

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WARN_IEEE_1500_ERROR_CODE/WARN_IEEE_1500_TIMEOUT_ERROR	0x81		0x01	
WARN_HBM_CORRECTABLE_ERROR	0x82			
WARN_HBM_UNCORRECTABLE_ERROR/WARN_HBM_TRAINING_FAILURE	0x83		0x01	
WARN_HBM_UNCORRECTABLE_ERROR/WARN_HBM_MBIST_FAILURE	0x83		0x02	
WARN_HBM_UNCORRECTABLE_ERROR/WARN_HBM_INVALID_PPR_ADDRESS	0x83		0x03	
WARN_HBM_UNCORRECTABLE_ERROR/WARN_HBM_PPR_FAILURE	0x83		0x04	
WARN_HBM_UNCORRECTABLE_ERROR/WARN_HBM_BASIC_MEMTEST_FAILURE	0x83		0x05	
WARN_HBM_UNCORRECTABLE_ERROR/WARN_HBM_ADVANCED_MEMTEST_FAILURE	0x83		0x06	
WARN_NVMCTRL_MEDIA_STATUS/WARN_NVMCTRL_MEDIA_NOTREADY	0x84		0x02	
WARN_NVMCTRL_MEDIA_STATUS/WARN_NVMCTRL_MEDIA_INERROR	0x84		0x03	
WARN_NVMCTRL_MEDIA_STATUS/WARN_NVMCTRL_MEDIA_RESERVED	0x84		0x04	
WARN_NVMCTRL_MEDIA_STATUS/WARN_NVMCTRL_MEDIA_DWR	0x84		0x05	
WARN_NVMCTRL_MEDIA_STATUS/WARN_NVMCTRL_MEDIA_AIT_NOTREADY	0x84		0x06	

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WARN_CFGCUR_STATUS/WARN_CFGCUR_SIGNATURE_MISMATCH	0x85		0x01	
WARN_CFGCUR_STATUS/WARN_CFGCUR_CHECKSUM_MISMATCH	0x85		0x02	
WARN_CFGCUR_STATUS/WARN_CFGCUR_REVISION_MISMATCH	0x85		0x03	
WARN_CFGCUR_STATUS/WARN_CFGCUR_DATASIZE_EXCEEDED	0x85		0x04	
WARN_CFGCUR_STATUS/WARN_CFGCUR_DATASIZE_OFFSET_EXCEEDED	0x85		0x05	
WARN_CFGIN_STATUS/WARN_CFGIN_SIGNATURE_MISMATCH	0x86		0x01	
WARN_CFGIN_STATUS/WARN_CFGIN_CHECKSUM_MISMATCH	0x86		0x02	
WARN_CFGIN_STATUS/WARN_CFGIN_REVISION_MISMATCH	0x86		0x03	
WARN_CFGIN_STATUS/WARN_CFGIN_DATASIZE_EXCEEDED	0x86		0x04	
WARN_CFGOUT_STATUS/WARN_CFGOUT_SIGNATURE_MISMATCH	0x87		0x01	
WARN_CFGOUT_STATUS/WARN_CFGOUT_CHECKSUM_MISMATCH	0x87		0x02	
WARN_CFGOUT_STATUS/WARN_CFGOUT_REVISION_MISMATCH	0x87		0x03	
WARN_CFGOUT_STATUS/WARN_CFGOUT_DATASIZE_EXCEEDED	0x87		0x04	

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版本：A 版次：0	内控等级：■I级 □II级 □III级

WARN_BIOS_CONFIG_HEADER_STATUS/WARN_BIOS_CONFIG_HEADER_CHECKSUM_MISMATCH	0x88		0x01	
WARN_BIOS_CONFIG_HEADER_STATUS/WARN_BIOS_CONFIG_HEADER_REVISION_MISMATCH	0x88		0x02	
WARN_BIOS_CONFIG_HEADER_STATUS/WARN_BIOS_CONFIG_HEADER_SIGNATURE_MISMATCH	0x88		0x03	
WARN_OS_CONFIG_HEADER_STATUS/WARN_OS_CONFIG_HEADER_CHECKSUM_MISMATCH	0x89		0x01	
WARN_OS_CONFIG_HEADER_STATUS/WARN_OS_CONFIG_HEADER_REVISION_MISMATCH	0x89		0x02	
WARN_OS_CONFIG_HEADER_STATUS/WARN_OS_CONFIG_HEADER_LENGTH_MISMATCH	0x89		0x03	
WARN_OS_CONFIG_HEADER_STATUS/WARN_OS_CONFIG_HEADER_SIGNATURE_MISMATCH	0x89		0x04	
WARN_NVDIMM_SURPRISE_CYCLE_STOP/WARN_POWER_CYCLE_POLICY_NOT_APPLY	0x8A		0x01	
WARN_NVDIMM_SURPRISE_CYCLE_STOP/WARN_POWER_CYCLE_POLICY_APPLY	0x8A		0x02	
WARN_FNV_ACCESS/WARN_INVALID_FNV_ACCESS_MODE	0x8B		0x01	
WARN_MEMORY_BOOT_HEALTH_CHECK/WARN_MEMORY_BOOT_HEALTH_CHECK_MASK_FAIL	0x8C		0x01	
WARN_MEMORY_BOOT_HEALTH_CHECK/WARN_MEMORY_BOOT_HEALTH_CHECK_CHANNEL	0x8C		0x02	

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L_MAP_OUT				
WARN_MEMORY_POWER_MANAGEMENT/WARN_MEMORY_PM_THERMAL_PMON_VR_NOT_FOUND	0x8D		0x01	
WARN_MEMORY_POWER_MANAGEMENT/WARN_MEMORY_PM_THERMAL_TABLE_NOT_FOUND	0x8D		0x02	
WARN_MEMORY_POWER_MANAGEMENT/WARN_MEMORY_PM_TEMPERATURE_THRESHOLD_RANGE_ERROR	0x8D		0x03	
ERR_MKTME_ENABLE/MINOR_ERR_SECURITY_POLICY_NOT_FOUND	0xE0		0x01	
ERR_MKTME_ENABLE/MINOR_ERR_PHYSICAL_ADDRESS_BITS_EXCEEDED_MAX	0xE0		0x02	
WARN_SILICON_CAPABILITY_LIMITATION/WARN_TME_ENABLED_CRYSTAL_RIDGE_NOT_SUPPORTED	0xE1		0x01	
WARN_SILICON_CAPABILITY_LIMITATION/WARN_SGX_ENABLED_CRYSTAL_RIDGE_NOT_SUPPORTED	0xE1		0x02	
WARN_SILICON_CAPABILITY_LIMITATION/WARN_TME_INTEGRITY_ENABLED_CRYSTAL_RIDGE_NOT_SUPPORTED	0xE1		0x03	
WARN_ROUNDTRIP_CALCULATION_RANGE_ERROR/WARN_DDR4_TO_DDR4_ROUNDTRIP_RANGE_ERROR	0xE2		0x01	
WARN_ROUNDTRIP_CALCULATION_RANGE_ERROR/WARN_DDR4_TO_DDRT_ROUNDTRIP_RANGE_ERROR	0xE2		0x02	

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ANGE_ERROR				
WARN_MINOR_WILDCARD	0xFF			
CPUPM POST Code				
STS_PPM_STRUCT_INIT	0xD0	Start of PPM structure initialization		
STS_PPM_CSR_PROGRAMMING	0xD1	PPM CSR programming		
STS_PPM_MSR_PROGRAMMING	0xD2	PPM MSR programming		
STS_PPM_PSTATE_TRANSITION	0xD3	Start of PState transition init		
STS_PPM_EXIT	0xD4	PPM exit		
STS_PPM_ON_READY_TO_BOOT	0xD5	PPM On ready to boot event		
IIO POST Code/IIO POST code - Major Definition{		IIO POST Code/IMinor Definitions		
STS_IIO_EARLY_INIT_ENTRY	0xE0	IIO early init entry		
STS_EARLY_PRELINK_TRAINING	0xE1	Early Pre-link training setting		
STS_GEN3_EQ_PROGRAMMING	0xE2	IIO Gen3 EQ programming		
STS_LINK_TRAINING	0xE3	IIO Link training		
STS_GEN3_OVERRIDE	0xE4	IIO Gen3 override		

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STS_IIO_EARLY_INIT_EXIT	0xE5	IIO early init exit		
STS_IIO_LATE_INIT_ENTRY	0xE6	IIO late init entry		
STS_IIO_PCIE_PORT_INIT	0xE7	PCIE port init		
STS_IIO_IOAPIC_INIT	0xE8	IOAPIC init		
STS_IIO_VTD_INIT	0xE9	VTD init		
STS_IIO_IOAT_INIT	0xEA	IOAT init		
STS_IIO_DFX_INIT	0xEB	IIO DFX init		
STS_IIO_NTB_INIT	0xEC	NTB init		
STS_IIO_SECURITY_INIT	0xED	Security init		
STS_IIO_LATE_INIT_EXIT	0xEE	IIO late init exit		
STS_IIO_ON_READY_TO_BOOT	0xEF	IIO On ready to boot		
STS_IIO_INIT			0x0	IIO On ready to boot
STS_IIO_MS2IOSF_CREDITS_ENTRY			0x10	IIO MS2IOSF Credit entry
STS_IIO_MS2IOSF_CREDITS_EXIT			0x1F	IIO MS2IOSF Credit exit
STS_IIO_MS2IOSF_BANK_DEC ODE_ENTRY			0x20	IIO MS2IOSF Bank decoder entry

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STS_IIO_MS2IOSF_BANK_DEC ODE_EXIT			0x2F	IIO MS2IOSF Bank decoder exit
ACPI Phase(OS Runtime)				
APIC	0xAA	OS invokes _PIC to notify ASL of current Interrupt controller mode	0x00	
PIC	0xAC		0x00	
	0x72	System sleep down	0x00	
	0x73	System Wake up	0x00	
	0xFB	Hot Plug Controller Command Method	0x00	
	0xFF	Eject Control Methods	0x00	
	Slot Numnber	PM_PME Wake Handler	0x00	
	0xFD	Check for Hot-Plug Events	0x00	
	0xFC	Hot-Plug Handler Exit	0x00	
	0x21-0x3F	required by WHQL but somehow SVOS breaks	0x00	
	0x4A		0x00	
ACPI	0xE3		0x00	
ACPI	0xEE		0x00	

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	0x01		0x00	
	0x62		0x00	
	0x84		0x00	
	0xFE	_L10 GPE Asserted	0x00	

6 发行与管制

本程序依《文件控制程序》制订、审核、批准与发行，其变更相同。